Executive Summary

Introduction

Aquatic physical therapy serves as an important way for patients to help improve strength, balance and range of motion, and wheelchairs that assist to transport patients into the water is an important part of that experience. Many of the current designs and wheelchairs in use have limited availability and have led to a dissatisfied user experience. Our team, Asians Don't Raisin, is working with Wave Therapies to attempt to provide an improved experience for both the patient and the physical therapist when using the wheelchair, creating a refined design that can be open-sourced, and potentially manufactured by other aquatic centers and therapists

Purpose

Many of the issues that have arisen stem from the introduction of a PVC pipe wheelchair, after production for the stainless steel version ceased. Issues include floating in the water, foot support, low durability, weak braking system, and multiple pain points along the frame of the wheelchair. Our improved design focuses on solving these problems, and aims for a lower-costing, highly durable, satisfying ergonomic experience for the patient being wheeled in as well as the therapist helping guide the wheelchair into the water.

Selected Design

Our design consists of a similar design, but with a few key changes to improve the user experience. First, we added the resting arms the same size for symmetry and a more familiar user experience. Rods will be added in order to increase the mass, adding to the weight in order to faster sink the wheelchair. One of the horizontal rods will be removed in order to save material and increase comfort while keeping similar structural integrity. The new mesh design holds stronger and will be bolted to the sides of the chair rather than on the edges of the seat, giving the user a better experience as those screws before caused injury when patients were attempting to get up from the chair. The brake will continue to be a scissor brake, however the location will be moved near the back of the wheelchair with multiple points of contact for increased durability and to dissuade users from leaning on it to get up, leading to inevitable part failure. The mag wheel will be the continued choice, a good airless option and the standard for wheelchair movement.

Implementation

Using PVC pipe for the frame, stainless steel rods for the added weight, and an improved mesh material for the patient to sit on, our design is structurally sound and provides added comfort compared to the older PVC pipe models. Overall, the cost for materials and manufacturing came out to $1100. The parts are manufactured so that any repair can be done relatively quickly. Functionality and load bearing was tested with full system testing, and mesh satisfaction was recorded through user survey.

Results

MECHANICAL ENGINEERING
UNIVERSITY of WASHINGTON

Capstone Design Project *Winter 2026*

Pool Access Wheelchair
Wave Therapies

Final Design Report Document

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1 Project Introduction

An aquatic therapy company, Wave Therapies, has hired us to design a wheelchair that can transfer their clients into the pool. They have requested that the wheelchair be comfortable, reliable, and easy to recreate.

1.1 Need Statement

The wheelchairs that our client currently uses are made out of PVC. While affordable and durable, PVC is lightweight and causes the wheelchair to float in the water, leaving users feeling unsupported during transfers. Wheelchair floating also causes some patients to slide/float out of the chair when entering the water. The seat material tends to wear out over time and is difficult to replace. Our client has one stainless steel wheelchair that they prefer to use, but the brakes tend to wear out due to patients using them as leverage when transferring out of the wheelchair.

1.2 Project Deliverables

Our end goal is to create a functioning prototype similar to the PVC wheelchairs the aquatic center currently uses while also addressing the main problems Wave Therapies has addressed, separated into the following categories: stability, structure, and comfort. We will develop several ideas for each problem and present them to our clients, working together to pick the best solution. We will combine these solutions into a working prototype and test it at Bellevue Aquatic Center's pool, adjusting our model until all are satisfied with the results.

1.3 Background and Significance

Individuals who experience physical disabilities such as chronic pain and neuromuscular disorders often cannot withstand the intensity of physical therapy on land. Instead, many turn to aquatic therapy, which can be more gentle on the body and allows the individual to use the water's natural support for balance. Aquatic therapy at Wave Therapies at the Bellevue Aquatic Center is maintained in a heated pool that eases joint pain, cushions falls, and uses water's natural resistance to make muscles work harder. Participants at Wave Therapies tend to enter the pool by wheelchair. Over time, the pool accessible wheelchairs break down, are too expensive to manufacture, uncomfortable for some individuals and sometimes make it harder for the physical therapists to use to transfer their patients into the pool.

1.4 Stakeholders

The highest priority stakeholder is, of course, the user. Those with disabilities rely on these wheelchairs on their physical therapy journey, and many of the current designs and products hinder their comfort and well-being. Many of the disabilities they face use aquatic therapy as it can be more comfortable for their situation, and the current wheelchairs do not align well for that purpose, so our improved wheelchair design will greatly benefit their aquatic therapy experience. Also affected is Wave Therapies, our client and the provider of those wheelchairs. Physical therapists need to ensure their patient is comfortable and responding positively throughout the session, and it is difficult to provide a comfortable atmosphere when the pool accessible wheelchair is faulty or causes great pain or discomfort. Our improved wheelchair design will consider all of the possible issues and create a product that can allow Wave Therapies' physical therapists to provide ample support to the various needs of their patients.

As the proprietor and owner of the aquatic facilities, the Bellevue Aquatic Center and therefore the city of Bellevue is an important stakeholder, as they need to ensure they can provide a proper establishment for its citizens to receive aquatic therapy to improve their physical wellbeing. If we were able to come up with a design that addresses all of the needs such as comfortability, reliability, and budget friendly, the aquatic center can take the blueprints to the city of Bellevue and manufacture them. Following positive reviews, they can then open source the chairs to different aquatic centers or physical therapy facilities.

2 Problem Statement

According to Wave Therapy the current model of pool accessible wheelchairs have issues with floating, comfortability, and not affordable. The PVC wheelchairs easily lose their balance due to buoyancy issues when they are being pushed down the ramp to the pool, and it takes them more than 3 minutes for the water to fill all the pipes and regain the balance. At the same time, the concern of comfortability is the locations of bolts on the mesh seat that can possibly cause pain for people with disabilities on their lower body as they lack muscles. Our team is focused on these issues to redesign the PVC wheelchairs in order to improve the user experience, and at the same time to design a more affordable model which has a high availability for components on the market if maintenance is required.

2.1 Customer Requirements

Since we agreed to take an original blueprint of one of the PVC wheelchairs that our client uses and improve on some features that prevented the product from being fulfilling for them, the top-level requirements for our primary customer are reliability and budget

friendly. Currently, the wheelchairs' braking system and seating break/wear down very easily. These features are material specific, making it difficult for the client to easily make repairs. Thus, we want to keep our design affordable and accessible as we want them to be able to replace any features on the wheelchair. The lower-level requirements are materials and adding another feature to the design. In terms of materials, our client did express that they favored a stainless steel wheelchair, however due to the timing and lack of budget, we are unable to design one that will be easy for them to replicate, so we will likely stick with PVC material. However, we did agree to research materials that are both durable and affordable, and will introduce it to our design if possible. Our clients also introduced a foot rest feature that some of the wheelchairs have, but they did state that they rarely use it for their clients.

2.2 Engineering Specifications

To fully understand the scope of this project, we must have engineering specifications that measure the critical to customer requirement so that we know that the customer's needs are met. Given in the table below, we outline the specification that each requirement must meet.

Critical to Customer Requirement	**Engineering Specification**	**Source**
Reliabile	Braking system withstands > 5 years of constant use	Client meeting
Budget Friendly	< $3000 (for stainless steel) < $2000 (for PVC)	Market prices for wheelchair types
Compact	< 0.8m wide (32")	Pool ramp measurement
Durable	350lbs weight limit	Client meeting

In addition to these critical to customer requirements, here are other requirements that this project contains. These requirements are not as important to the customer, but will help with the maintenance and enjoyment of the wheelchair.

Requirement	**Engineering Specification**	**Source**
Comfortable	Not slippery for user (tested for 1 month without issues) Easy on the weaker muscles	Client meeting

	of clients (tested for 1 month without issues)	
Non Floating	< 2 mins before stops floating	Current time that wheelchairs take to stop floating
Weight ***TBD depends on final design and prototype testing.***	Ideally, the chair weight and extra water/ extra steel should be greater than the buoyancy force, ~54 lbf, calculated in *Feasibility Analysis.*	Previous chair weight, making it easier to pull out of the water
Mesh Easily Replaceable	Able to replace the mesh with basic tools in < 30min	Client meeting
Able to Withstand Environment	Lasts > 5 years in warm, humid, chlorine environment	Typically, the old versions would degrade this quick

A.1 Codes and Standards

ANSI/RESNA WC series - standard for wheelchair design and testing. More specifically, we are focused on the manual wheelchair aspects of this standard. This would be WC-1, WC-3, WC-4, WC-5, and WC-8. These standards are meant to control the performance and testing (WC-1), seating system (WC-3), manual propulsion and casters (WC-4), dimensions and clearances (WC-5), and durability and fatigue (WC-8).

2.3 Design trade-offs and challenges

The four main aspects we wanted to focus on in our new design were comfortability, structural stability, anti-tipping and sinking time. However, since this product is a wheelchair that is being used by the city, it had to fit the city's standards, which we could not find, so we tried making the wheelchair as similar to the original wheelchair our clients were using.

To combat the discomfort patients feel on the original wheelchair, we removed the two horizontal bars on the seating area of the original wheelchair shown in Figure 2.3.1. These components were identified as primary sources of discomfort, largely due to the reduced muscle mass commonly observed in the patient population compared with the general

population. Because soft tissue provides natural cushioning during seated activities, insufficient muscle mass results in increased pressure and direct contact between the support bar and the patients' lower extremities. Consequently, patients frequently reported discomfort from the bar pressing against their legs, and the design also made transfers and exiting the chair more difficult. However, after removing the two horizontal bars, we realized that they were also used to keep the frame of the wheelchair from caving inwards when the patient sat down. To combat all of these factors, we created a criss-cross structure on the bottom of the wheelchair, below the mesh, in the middle of the seating area. This way, the patient would not feel the bars, and the wheelchair would still be structurally stable. We also made the back bar of the wheelchair curved and added a pool noodle to maximize comfort.

To combat the tipping issue that the original wheelchair had, we first calculated where the point of rotation acts on the wheelchair, which was where the wheels were at, and we calculated and compared the weight of the wheelchair itself and the wheelchair with water filled in it, to see how much more weight we would need to add onto the wheelchair. And with budget restraints, we calculated that we would have to drill holes on the wheelchair and add steel ballast rods to reverse the tipping issue. Originally, we wanted to use stainless steel, however that is more expensive, so we decided to cover the steel ballast rods with Pasti-Dip which will elongate the time of corrosion.

In terms of sinking time, our clients’ original wheelchair took around 2 minutes to fully sink into the water, so to reduce the time, we decided to drill holes on the bottom area of the wheelchair and the arm rests to fill the wheelchair with water quickly and resulted in reducing the sinking time by 80%.



Figure 2.3.1: Original wheelchair that clients are currently using to transfer their patients from land into the pool. The two horizontal bars circled in red were significant discomfort factors.

3 Conceptual Design, Evaluation, Selection

Based on the needs that were listed in section 2, we have started designing our wheelchair by first creating a replica of the current PVC wheelchair design that the aquatic center is using. By doing so, we were able to calculate the weight and buoyancy force of the wheelchair when it is fully submerged in the water. With these calculations, we were able to start researching materials such as the type of PVC that will be used, stainless steel rods, and the mesh all while budgeting to make sure it doesn't surpass the average price of a pool wheelchair.



Figure 3.0.1: Full CAD model depicting the 0th prototype

3.1 Project Decomposition

To ensure a successful project, we broke down each of the main wheelchair systems and the main things we are looking to improve: frame, mesh, wheels, and brake. Shown below are the main project member functions, however we will each collaborate to work on these systems, looking for improvements, go through new calculations, and prototype and test these new ideas.

Project manager: Katie Leija

- Track project's health, milestone completion
- Assign tasks to team members
- Design

Document Lead: Kit Nguyen

- Track documents, budget, meeting notes
- Design

Technical Lead: Hoang (Long) Dinh, Christopher Tran

- Research and select materials
- Design and Ensure the prototype meets of ISO 7176 and ANSI/RESNA standards

Customer Liaison: Shannon Kwong

- Communicate with clients
- Design and Evaluate

3.2 Concept Generation

The main redesign category is divided into four groups according to four main subfunctions: frame, mesh, brakes, and wheels. Ideation methods used to generate concepts for the subfunctions include internet researching, dismantling the given wheelchairs, and utilizing Morphological charts. By physically dismantling the existing stainless steel wheelchairs, the team measured the general dimensions of a wheelchair and also explored how to install the wheels. Concurrent research provided suitable materials for PVC frames and mesh fabric that have high chlorine resistance. These individual findings or self-idea were then integrated into a Morphological chart, allowing the team to systematically mix and match sub-components to generate diverse design concepts.

- **Subsystem 1: Frame**



Figure 3.2.1: Schedule 40 PVC

To ensure the detailed design process could begin immediately, an extensive material trade study was conducted at the project's inception. As detailed in the Engineering Specifications matrix (Figure D.1), the team evaluated multiple polymers and metals, including HDPE, Polypropylene, Aluminum, Stainless Steel, and various PVC grades, across critical parameters. Because the wheelchair must operate reliably in a submerged aquatic environment, chlorine resistance was established as a mandatory criterion, which immediately eliminated 5052-H32 Aluminum. The remaining candidates were then optimized for yield strength, elastic modulus, specific gravity, and budget constraints. Heavy Duty Schedule 40 PVC was ultimately selected as the primary frame material; it provides excellent structural rigidity with a yield strength of 7,000 psi, inherent chlorine resistance, and superior cost-efficiency ($1.44/ft) compared to specialty polymers like PVDF or heavier metallic alternatives.

To permanently bond the Schedule 40 PVC components, the team selected a two-stage solvent welding technique using an industrial PVC primer and solvent cement. Unlike mechanical fasteners that can corrode or loosen under dynamic loading, solvent welding chemically fuses the pipes and fittings into a single continuous structure. This method was specifically verified for the aquatic environment, as the cured solvent joints are entirely waterproof, highly resistant to pool chlorine, and completely safe for prolonged submerged use in a clinical facility.

- **Subsystem 2: Mesh/ Fabric**

The team wanted to ensure that the mesh would be durable, comfortable and easy to access when it was needed to replace it. We researched different types of mesh materials and came across a website that sold patio furniture and found that they had a wide variety of different meshes of patterns and textures. Through this research, we learned that PVC Olefin was most durable in chlorinated water and would also be comfortable for patients to sit.

With all of this data, we were able to pinpoint one mesh that was durable in terms of longevity, easy to drain, and comfortable for patients. The next decision we had to make was how we would put the mesh on the wheelchair. On the original wheelchair our patients were using, the mesh was bolted on the seat area, which clients often complained about saying that their legs would bump into the bolts of the mesh causing discomfort. We then decided to bolt the mesh on the side of the seating area, therefore it would be easy to remove the mesh if needed, and it would not intrude on the patients’ comfortability.

- **Subsystem 3: Brakes**

In attempting to come up with a new idea for the brake, we looked at it from two different angles: how we could alter the current brake to work better for the client’s needs and if a

different type of brake could be used for higher satisfaction. To this end, we came up with ideas for both scenarios, and compared the expected success of the idea with how easy it would be to implement/purchase the part for. For each idea we came up with a pros and cons list to compare the ideas and received feedback from both our client and mentor to determine which idea would be best. We then discussed as a group which possible brake option would be best, and moved forward with that design.

- **Subsystem 4: Wheels**

Similar to the brake subsystem, we looked at both different types of wheels that could potentially be used for this project, as well as what changes we could make to the current wheel type in order to better suit the client's needs. We again discussed the pros and cons of each idea, presented it to our client/mentor, and compared each idea through group discussion and a morphological chart to determine which wheel to move forward with.

3.3 Conceptual Design

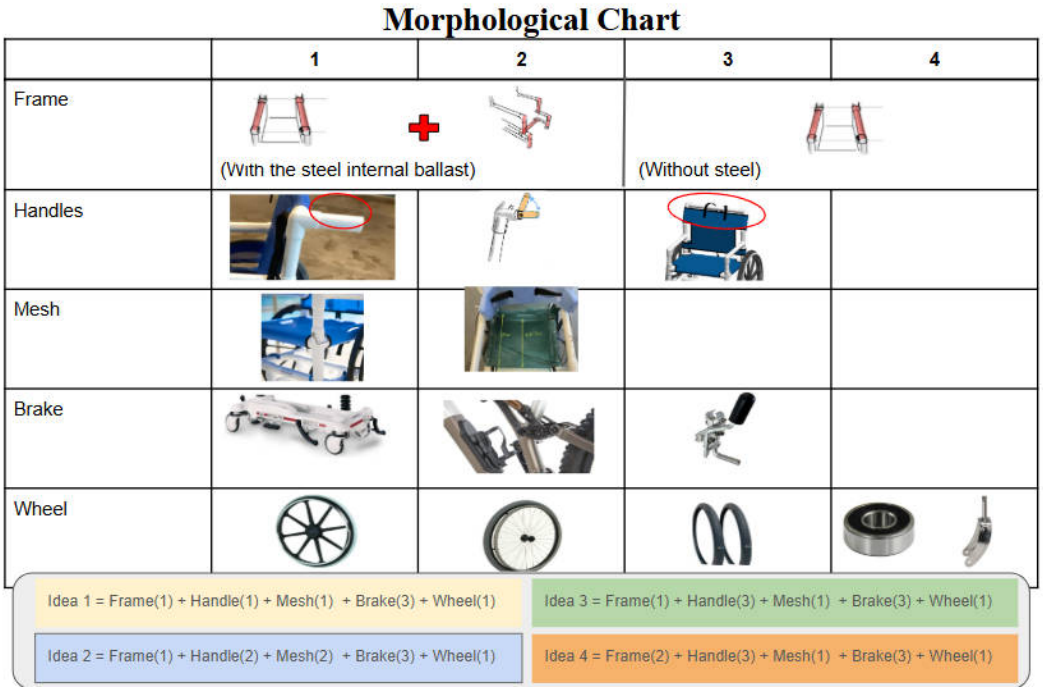


Figure 3.3.1: Morphological chart of four ideas

As we mentioned above, a Morphological chart was used to combine all the ideas of the sub-system together. The figure above shows four conceptual designs that the team came up with for MileStone 2.

3.4 Feasibility Analysis

To assess the feasibility of the selected concept, the team developed parametric 3D models using Onshape CAD software. These digital assemblies were utilized to extract precise mass properties and to analyze the distribution of forces acting on the wheelchair under various operational load cases.

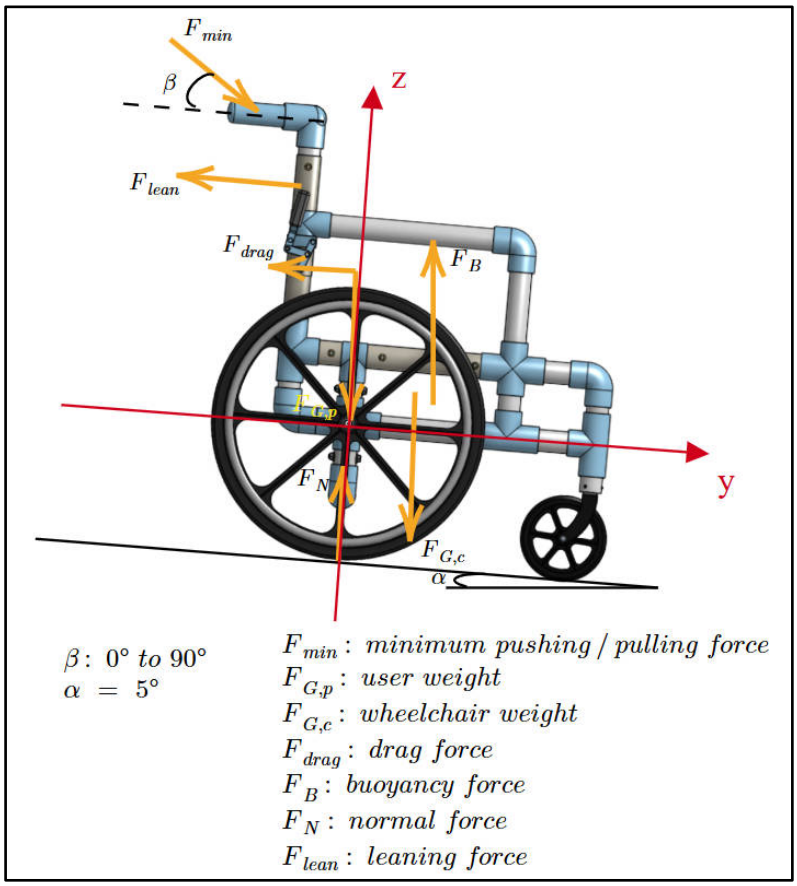


Figure 3.4.1: FBD for equilibrium moment analysis

The formulas we were mainly using in this analysis are:

Buoyancy force: $F_B = V_{displayed} \cdot \rho$ (Calculating using IP units)

Drag force: $F_D = \frac{1}{2} C_D A \rho v^2$

Rotational moment: $M = F \cdot d$,

This static moment analysis was performed to determine the minimum pushing or pulling force required to counteract tipping phenomena and to define the minimum internal ballast mass necessary to resolve the wheelchair's observed positive buoyancy. **(see *Appendix - C*)**.

3.5 Concept Evaluation

To determine the most optimal designs from the Morphological chart, the team evaluated each design by applying the weighted decision matrix. The ideal selections prioritized the unique constraints: sinking ability, comfortability, durability, and cost. For each criterion, the weight from 1 to 5 point, with 1 was the lowest score and 5 was the highest score.

Decision Matrix:

Ranking point: 1 (bad, lowest) to 5 (good, highest)

	Sinking	Weight	Comfortability		Durability				Cost	Total Score
			Frame + Pushing Handle	Mesh	Frame + Pushing Handle	Mesh	Brake	Wheel		
Idea 1	TBD	Slightly different with different type of pushing handle 4	4	4	5	3	3	4	4	31
Idea 2			3	2	3	5	3	4	3	27
Idea 3			2	4	3	3	3	4	4	27
Idea 4		2	2	4	3	3	3	4	5	26

Figure 3.5.1: Decision matrix table from Milestone 2

3.6 Concept Selection

The two most promising designs are:

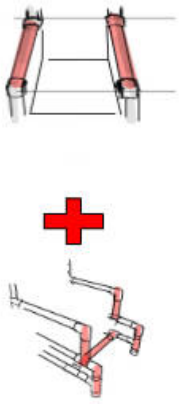
Frame (with steel)	Handle	Mesh	Brake	Wheel
				

Figure 3.6.1: Design 1-Frame adjustment, internal ballast, and wrap-around mesh (Idea 1)

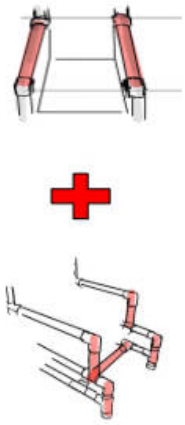
Frame (with steel)	Handle	Mesh	Brake	Wheel
				

Figure 3.6.2: Design 2-Frame adjustment, internal ballast, wrap-around mesh, no handle (Idea 3)

These temporarily finalized conceptual designs consist of the following integrated sub-system: SCH 80 PVC/ SCH 40 PVC, PVC Olefin/ PVC Olefin Plus/ PVC Plus Mesh, stop brake, and mag wheel. The heavy duty PVC types provide a good yield strength for structural integrity. The seat and backrest utilize a high tension, 100% vinyl-coated polyester or vinyl coated polyester with Olefin blend mesh from Phifertex brand. This provides high dimensional stability (preventing sag under the user's weight) and features an open weave to ensure instant water drainage upon exiting the pool. The stop brake is selected because of its simplicity in design, functionality, and installation. Lastly, the mag wheels are the industry standards for submersible and aquatic wheelchairs primarily because of their corrosion resistance and structural durability. ***(see Appendix - D for more materials specifications).***

While the Morphological chart details four primary concepts currently under evaluation, the team recognizes that design is a highly iterative process. The two most optimal selections are also temporary for us to move on with the next steps. However, we are actively continuing to research new redesign opportunities and remain open to integrating more concepts as the project evolves and more testings are run. For example, the internal ballast idea is still under research. The 316L Stainless Steel was the most suitable candidate thanks to its high chlorine resistance, but it is very expensive compared to 4140 Steel, which is not chlorine resistant. Therefore, the team is looking for a method to seal the internal ballast completely, which would open more options for us to also consider using non-chlorine resistant materials, such as 4140 steel as we have mentioned.

4 Preliminary Design

For our preliminary design, we focused on the factors in our conceptual design: the frame, mesh and weight. In terms of the frame, we took the feedback that was given to us about the elements that are uncomfortable, such as the front two bars where the client's patient's legs make contact with. For the mesh, our client only spoke about the need for longevity of the condition of the mesh, which made us research other types of meshes and research other areas where to install the mesh to work on the comfortability aspect. And in terms of weight, our client preferred that the wheelchair would sink faster and not tip backwards when the client leans backwards, so we calculated the weight of our new wheelchair design, and thought about different places to place steel rods in the wheelchair.

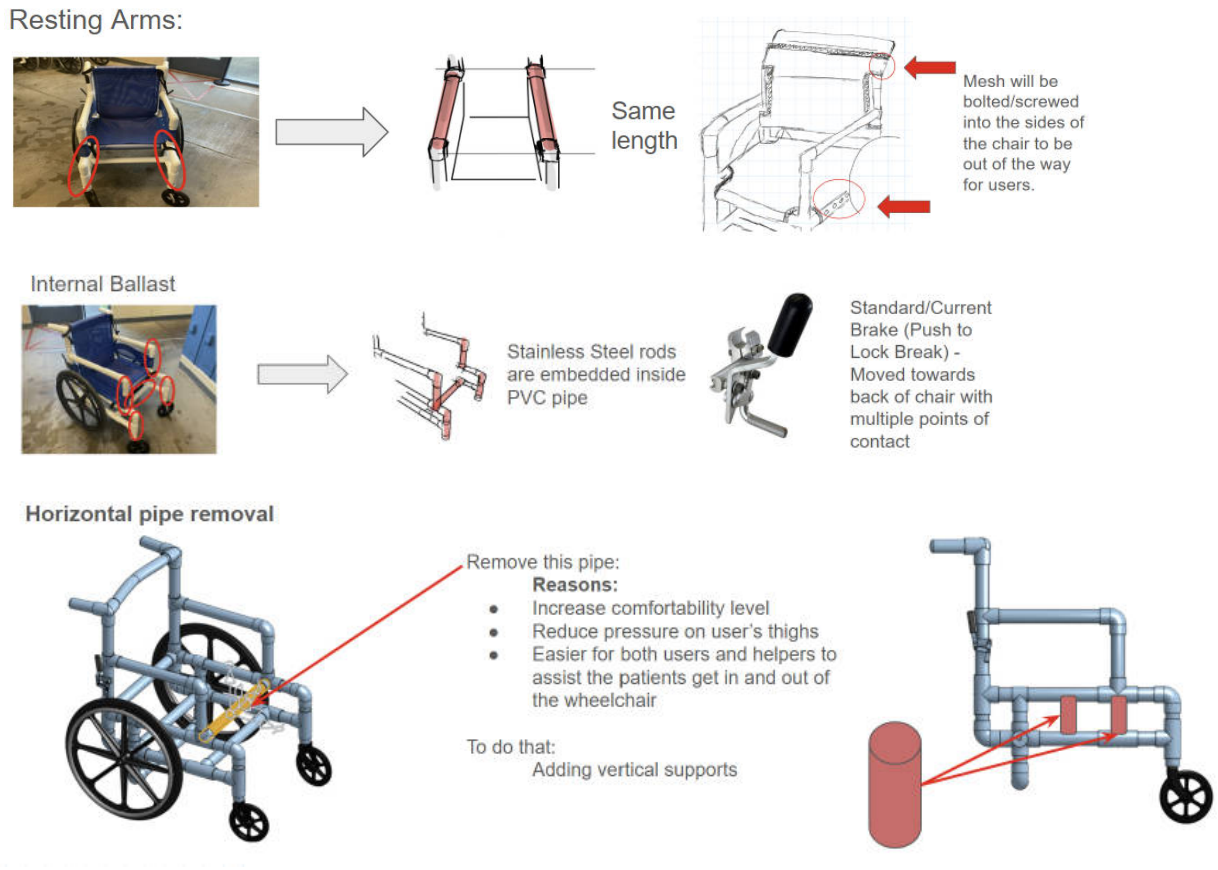


Figure 4.0.1: Preliminary design sketches and plans

4.1 Embodied Design

For our embodied design, our main factor that we decided first was to choose what material to use, since it will affect the weight and structure of the wheelchair. Since we wanted a heavy and durable material, we chose PVC Schedule 80, as it is durable in the aquatic environment, does not break easily and is thick, so it will add weight to the wheelchair. For mesh, we were able to research different types of meshes, specifically PVC blends which differ in how much space is in between the threads. And in terms of frame, we decided to make the sitting area smaller by removing the two front horizontal pieces of the frame near the leg area to maximize comfortability, and added supports to the frame by adding vertical bars attaching the bottom pipes of the frame to the pipes creating the seating area.

For our embodied design, we first focused on choosing the material for the frame, as this will affect the weight, structure, and strength of the wheelchair. We chose to use PVC Schedule 80 for the final product, as it is durable and resistant to chlorinated water. For the mesh, we researched different fabrics and decided upon a PVC coated polyester mesh. We then ordered

free samples of different blends and weaves and tested their chlorine resistance and user preference. After testing, our clients picked a woven vinyl-coated polyester and olefin blend (PVC/Olefin Blend) due to its durability and porosity.

4.2 Application of Codes and Standards

When creating this design, we made sure that each change we were adding followed relevant pool wheelchair codes and standards. In terms of changing the frame, we made sure to follow the ADA (Americans with Disabilities Act) which focuses on maintaining accessibility for individuals with mobility impairments in aquatic environments, specifically the ADA 2010 Standards- Section 1009 which requires that swimming pools provide one accessible means of entry such as a transfer system. This relates to the frame of the wheelchair as the seat height has to be compatible with the pool deck, has to have sufficient structural strength to support the patients during movement and transfer, and must have corrosion-resistant materials suitable for wet environments. The ISO also has ISO 7176 series that define testing methods for strength, durability, stability and braking performance that also helps reassure the measurements and material of the wheelchair. Mesh materials were also chosen from pool chair companies to ensure that the material was durable for wet environments and could last long in any condition and were easy to clean. The ASTM D5034 (American Society for Testing and Materials) provides a tensile strength test for textile fabrics, ensuring the mesh material can support the weight of the user without tearing or permanently deforming. In terms of chemical resistance, ASTM D543 is used to determine how the polymers and coated fabrics react to chemical exposure, including chlorine and pool cleaning agents.

4.3 Modeling and Analysis

Modeling and analysis were conducted to evaluate the structural performance and stability of the pool wheelchair design. The analysis involved calculating moment forces, hydrostatic buoyancy during full submersion, and the mass properties of the wheelchair across various stainless steel ballast configurations. This evaluation is categorized into three primary studies: Sinking Analysis, Tipping Analysis, and Load Capacity Analysis. Detailed mathematical models and calculations are provided in **Appendix C**.

1. **Chair weight versus the maximum buoyancy force when the chair is fully submerged:**



Figure 4.3.1: 5 Proposed Locations for Internal Stainless Steel Ballasts

To evaluate the submersion characteristics of the initial prototype **(Figure 1),** Onshape's mass properties tool was utilized to extract the total dry weight and the total material volume of the frame. Assuming a steady-state flooded condition (100% water ingress), the displaced volume of the PVC shell is 1047.439 in^3, which yields a maximum upward buoyancy force of 37.81 lbs. Because the frame's total weight of 47.361 lbs exceeds this buoyant force, static analysis confirms the wheelchair is inherently negatively buoyant. However, this net downward force of approximately 10 lbs is insufficient to ensure a rapid descent, potentially replicating the prolonged "floating" transition phase observed with the current chairs at the Bellevue Aquatic Center. To achieve a target submersion time of under one minute, the addition of internal stainless steel ballasts is required. While five potential ballast locations were identified during the **Preliminary Design** phase, the subsequent kinematic tipping analysis will determine the exact quantity and optimal placement of these weights. **(See Appendix C).**

Note: *It is important to note that the forces and masses calculated in this feasibility analysis represent an early stage theoretical baseline. In the physical prototype, slight numerical variations are expected due to manufacturing tolerances, material density imperfections in the PVC, and precise variations in the commercial wheel weights. Furthermore, during the final assembly phase, a structural cross-support system was integrated beneath the seat to resolve a frame"caving in" issue and increase overall frame integrity. While this additional sub-structure was not included in the initial kinematic model, its forward placement shifts the wheelchair's center of gravity, effectively increasing the stabilizing clockwise (CW) moment. Consequently, the original mathematical model presented here serves as a highly conservative worst-case scenario. The core tipping dynamics remain entirely valid, and the final physical wheelchair inherently possesses a higher margin of safety against rearward tipping than these baseline calculations suggest .*

2. **The minimum amount of pushing/ pulling force required to counteract the tipping issue when approximately half of the chair is submerged:**

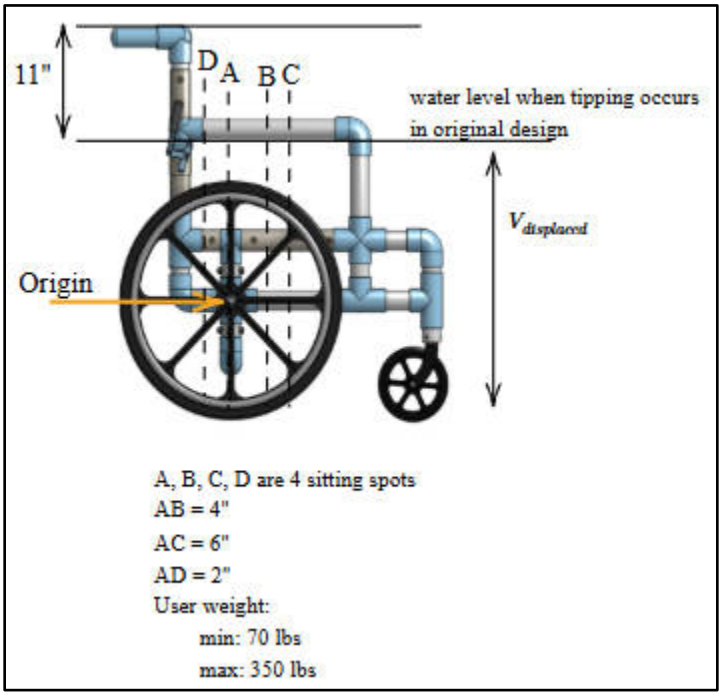


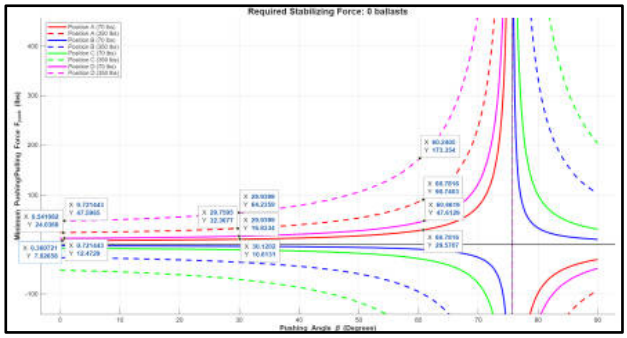
Figure 4.3.2: FBD for tipping analysis

To evaluate the wheelchair's tipping hazard during ramp entry, a moment equilibrium analysis was conducted at the point of rotation (rear wheel's center). Based on the physical testing on the existing aquatic center chairs, the critical water level inducing maximum tipping potential was established at the armrest height.

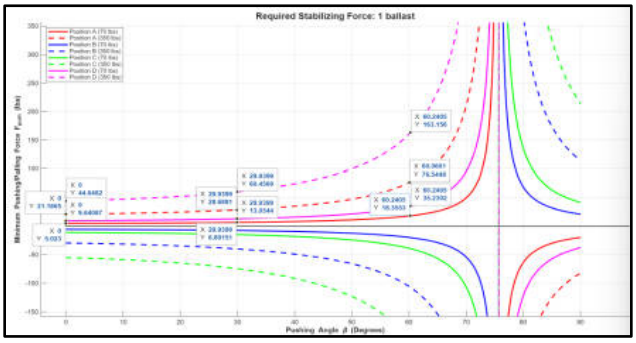
The simulation calculated the minimum required force at the therapist's handle across applied ranging from 0° to 90°. To capture a comprehensive envelope of operational scenarios, the model was analyzed under four distinct user seating positions (denoted as locations A, B, C, and D) and a user weight range from 70lbs to 350lbs. Location D was specifically modeled to account for the "hammocking effect" of fabric seats under heavy loads. To optimize stability without introducing excessive dry weight, eight parametric ballast configurations (ranging from 0 to 5 internal ballasts) were analyzed.

Results and Discussion:

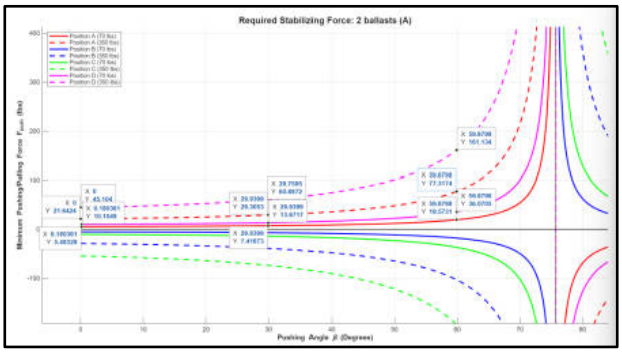
*(**Note**: For higher-resolution versions of the performance figures and the complete, step-by-step mathematical derivations supporting this analysis, please refer to the full analytical report provided in Appendix C).*



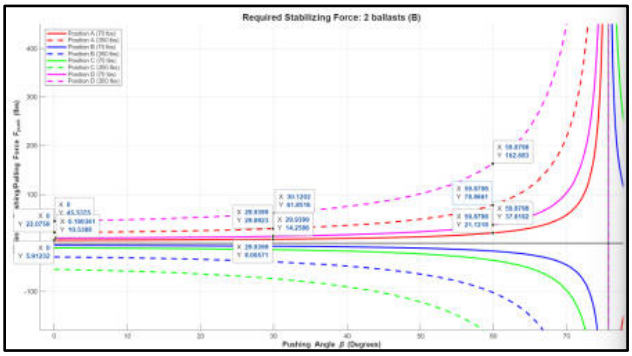
a) 0 ballast



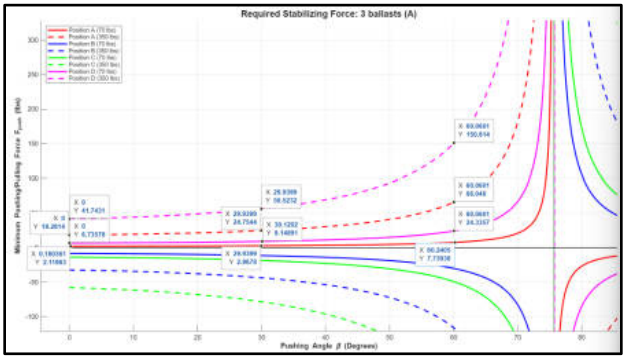
b) 1 ballast



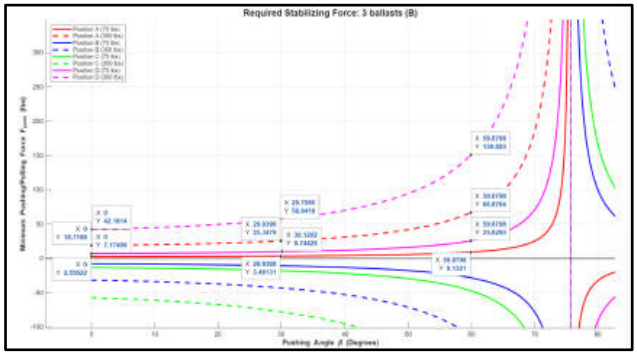
c) 2 ballasts (A)



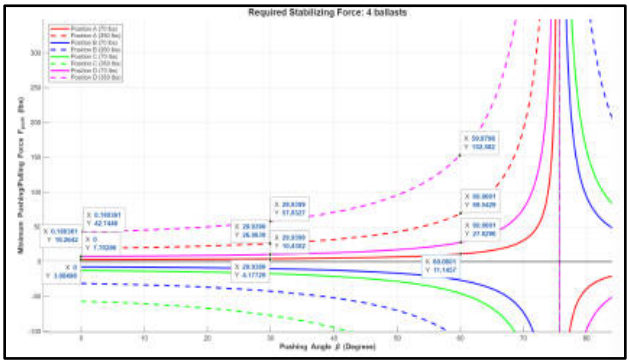
d) 2 ballasts (B)



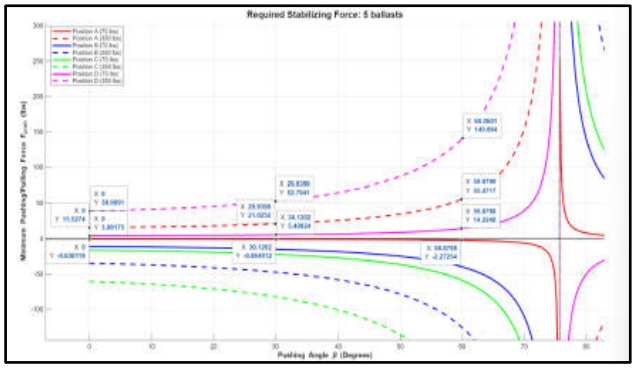
e) 3 ballasts (A)



f) 3 ballasts (B)



g) 4 ballasts



h) 5 ballasts

Figure 4.3.3: Plots of minimum pushing force vs pushing angle for 8 different configurations

Across all configurations, users seated in forward or neutral positions (Locations B and C) generate a stabilizing clockwise moment that dominates the tipping (counterclockwise moment). In these scenarios, the wheelchair exhibits inherent rotational stability and requires zero corrective pushing force from the therapists or helpers.

Tipping hazard only emerges when the user's center of gravity shifts significantly rearward, either by sitting at the extreme rear of the seat (location A) or due to severe fabric deflection (location D). Under these conditions, if the attendant applies force at an angle greater than 60°, the required stabilizing force spikes exponentially, exceeding 150lbs for a 350lbs user. This analysis mathematically confirms that therapists/ helpers must practice operating/ pushing the handle at a low angle (ideally between 0° and 30°) to safely control the wheelchair.

Crucially, the tipping analysis demonstrates that the required stabilizing force is practically independent of the internal ballast layout. Comparing the worst case scenario (location D, 350lbs user, 30° pushing force direction), increasing from 0 ballast to a maximum of 5 ballasts only reduces the required pushing force by 11.35 lbs. This marginal ergonomic improvement comes at the cost of increasing the total dry weight up to 72.6 lbs.

This decoupling of design variables allows the team to prioritize ballast configurations, such as 2-ballast or 3-ballast layouts, that optimize the submersion speed while minimizing the final dry weight and overall manufacturing cost.

3. Load Capacity Analysis:

A static structural analysis was performed using ANSYS Workbench to evaluate the total deformation and equivalent von Mises stress of the original PVC wheelchair frame, as well

as 5 other possible configurations under a maximum static user load of 350 lbs, in order to determine which design would provide enough structural support, in order to increase the comfort by removing the original horizontal bar. To ensure computational accuracy and result convergence, a mesh sensitivity study was conducted. The global element size was iteratively refined from the default automatic sizing down to 0.50 inches in increments of 0.1 inches, all the way to 0.1 inches. This gave us a mesh size of 0.2 inches to work with as the refined element size, as it balances out computation runtime with model accuracy. This refinement strategy was critical for improving the overall mesh metric quality; specifically, it minimized the quantity of highly distorted elements with a skewness greater than 0.80, which can artificially over-stiffen the mathematical model and compromise the reliability of the localized stress results. At a refined global mesh size of 0.2 inches, highly skewed elements were sparsely distributed and safely isolated from critical load-bearing regions. The element method was also changed to be quadratic, and using tetrahedrons as the element shape, in order to have a better element shape with which to capture all of the correct equivalent stress and total deformation throughout each configuration model. Configuration A consisted of a horizontal bar in between the two horizontal levels of the PVC seating frame structure, with it placed more in the middle of where the user will sit. Configuration B is similar to Configuration A, but rather than being connected at both levels, it is connected only at the bottom level of the seating frame. Configuration C had 45 degree elbows connecting diagonal crossing pipes in between the two horizontal levels. Configuration D and E are similar in that they each had sling tees rather than 45 degree elbows, and still had the diagonal crossing pipes across both levels as the support structure, however Configuration E used smaller diameter pipes as the cross in the hopes of bringing even more comfort to the user, as the smaller pipes would get even less in the way of the user. Total deformation and equivalent stress images for each model with a load of 200 lb were taken and recorded, down below. **(see Figures … below).**

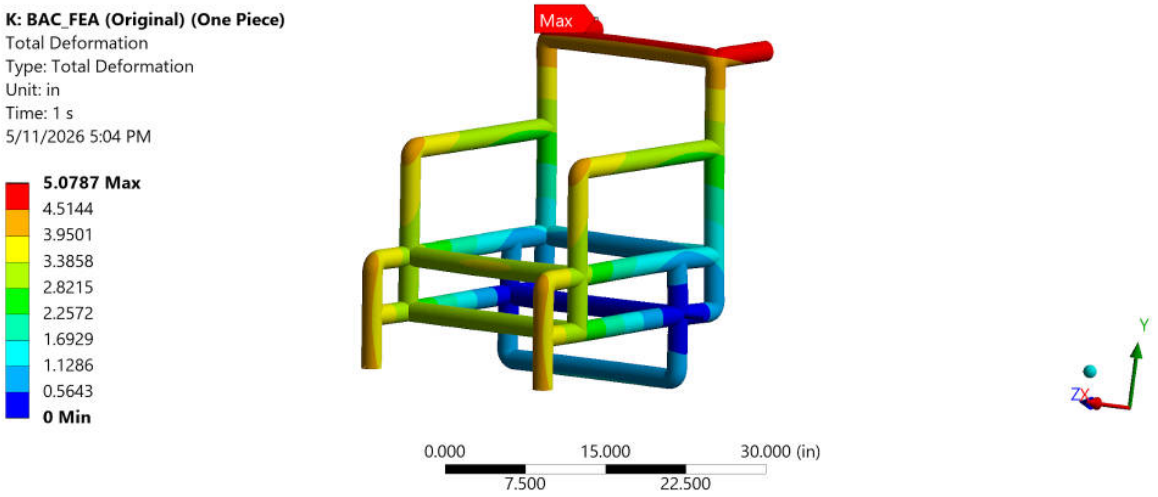


Figure 4.3.4: Total Deformation for Original PVC design for 200 lb load at mesh size 0.2 in

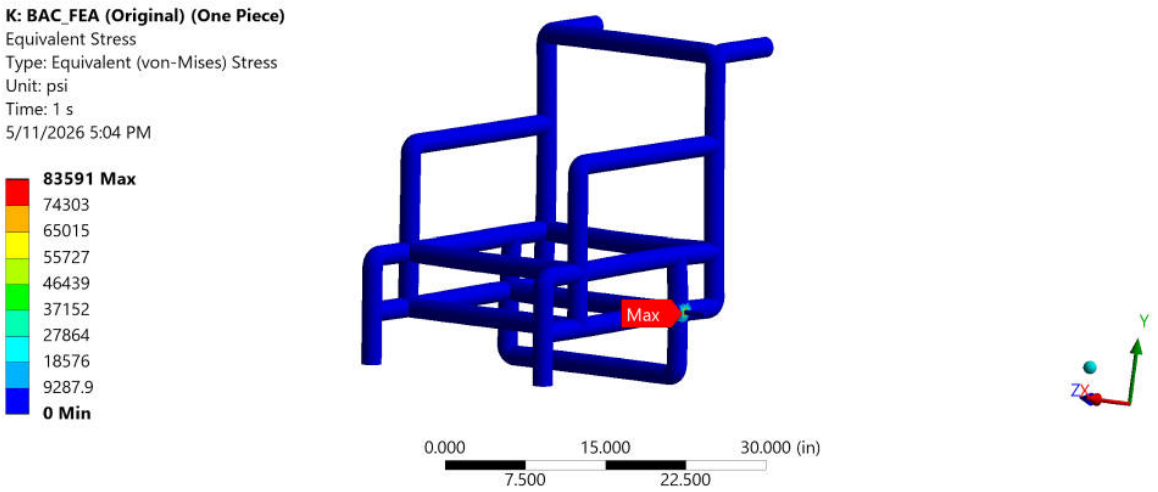


Figure 4.3.5: Equivalent Stress for Original PVC design for 200 lb load at mesh size 0.2 in

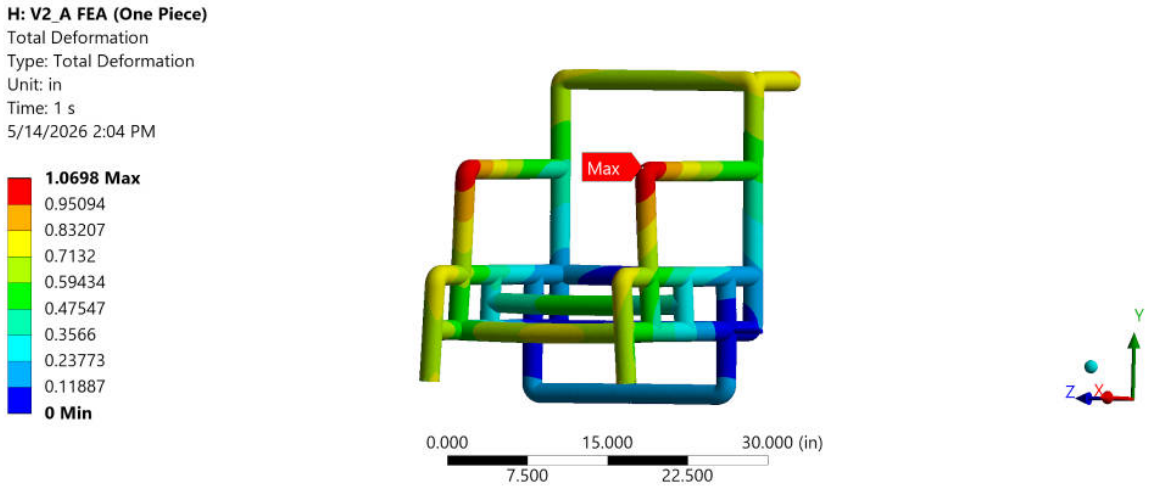


Figure 4.3.6: Total Deformation for Configuration A design for 200 lb load at mesh size 0.2 in

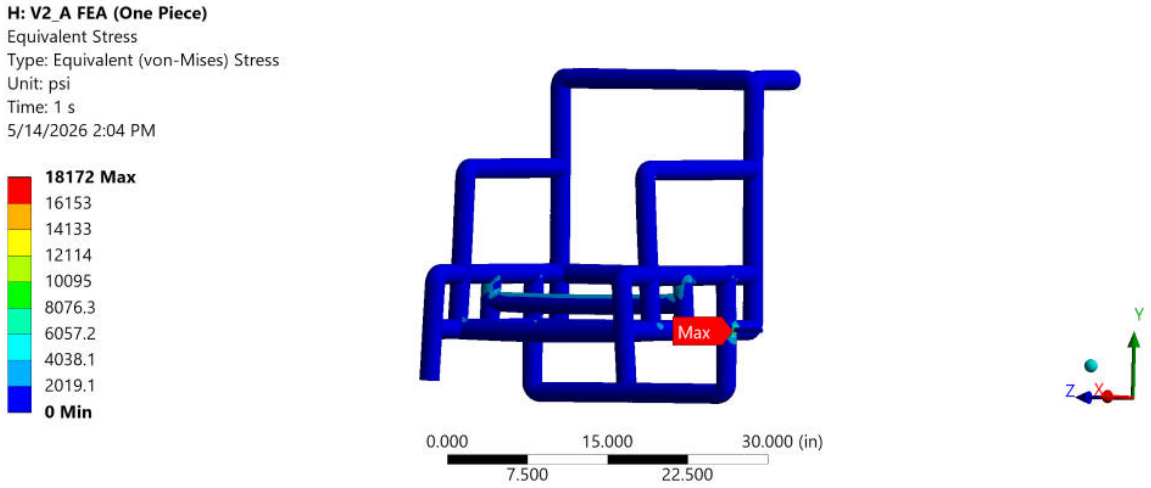


Figure 4.3.7: Equivalent Stress for Configuration A design for 200 lb load at mesh size 0.2 in

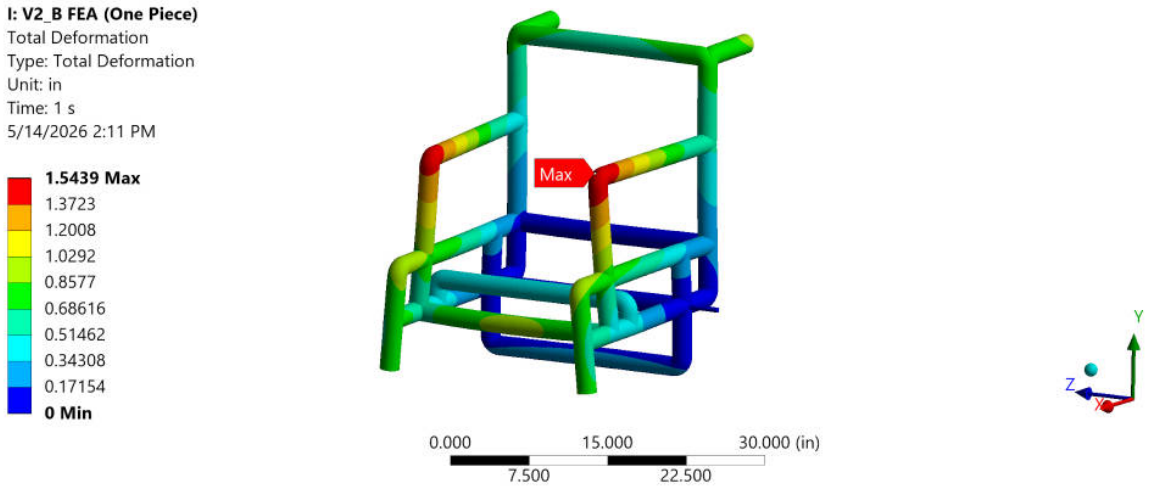


Figure 4.3.8: Total Deformation for Configuration B design for 200 lb load at mesh size 0.2 in

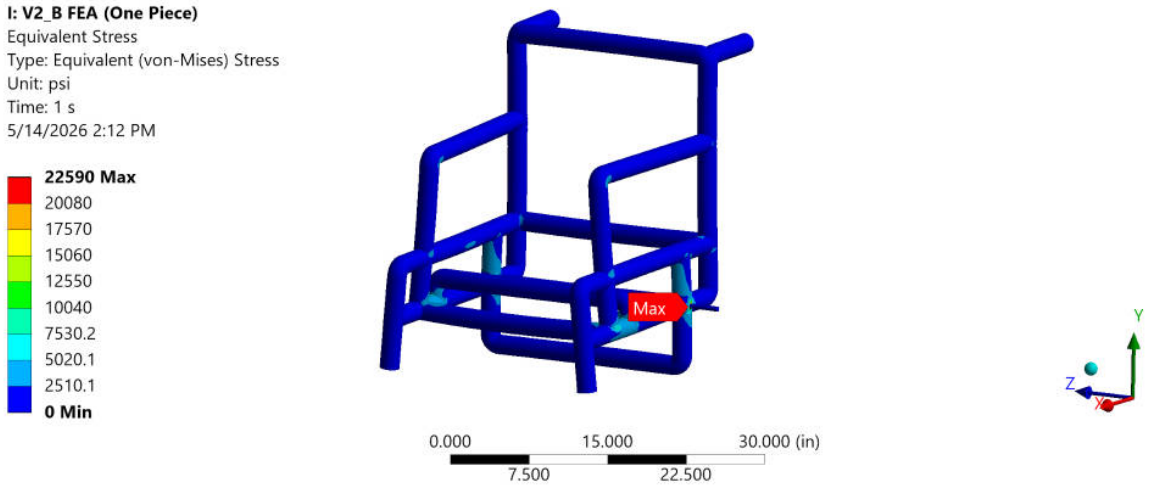


Figure 4.3.9: Equivalent Stress for Configuration B design for 200 lb load at mesh size 0.2 in

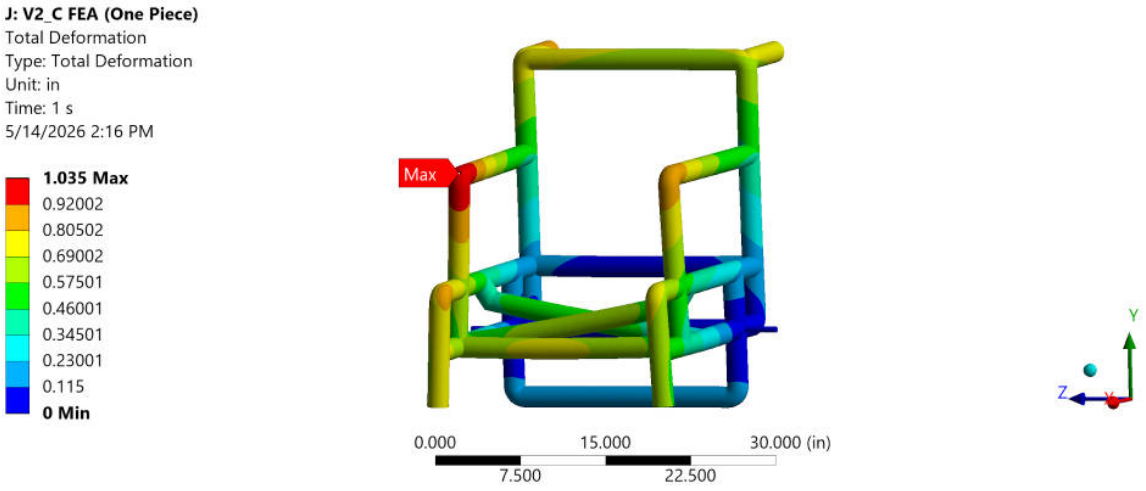


Figure 4.3.10: Total Deformation for Configuration C design for 200 lb load at mesh size 0.2 in

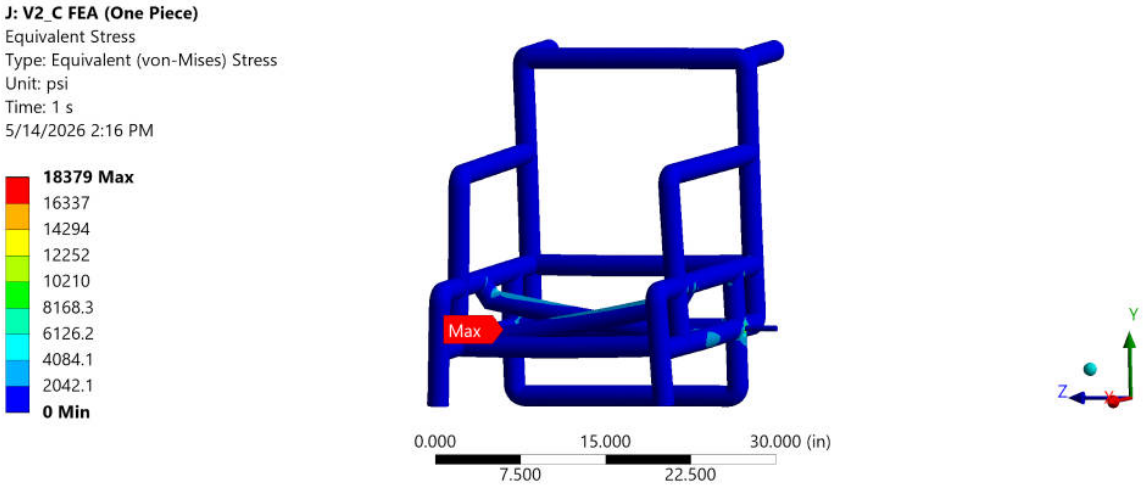


Figure 4.3.11: Equivalent Stress for Configuration C design for 200 lb load at mesh size 0.2 in

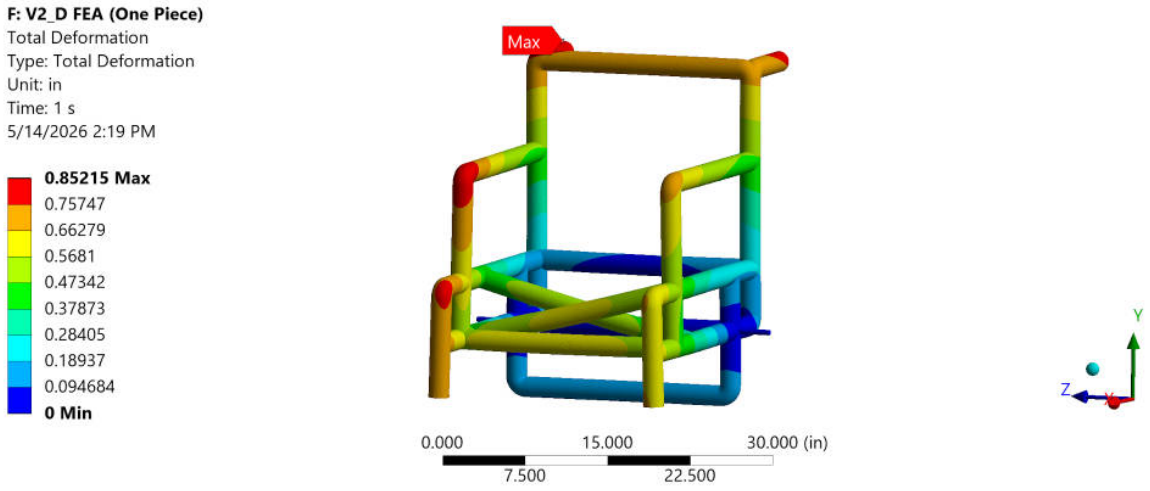


Figure 4.3.12: Total Deformation for Configuration D design for 200 lb load at mesh size 0.2 in

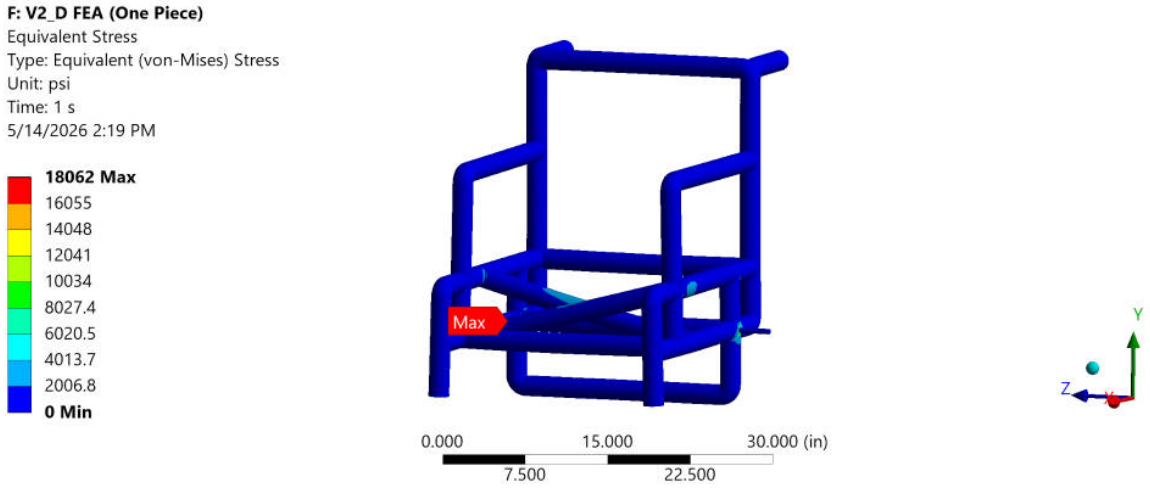


Figure 4.3.13: Equivalent Stress for Configuration D design for 200 lb load at mesh size 0.2 in

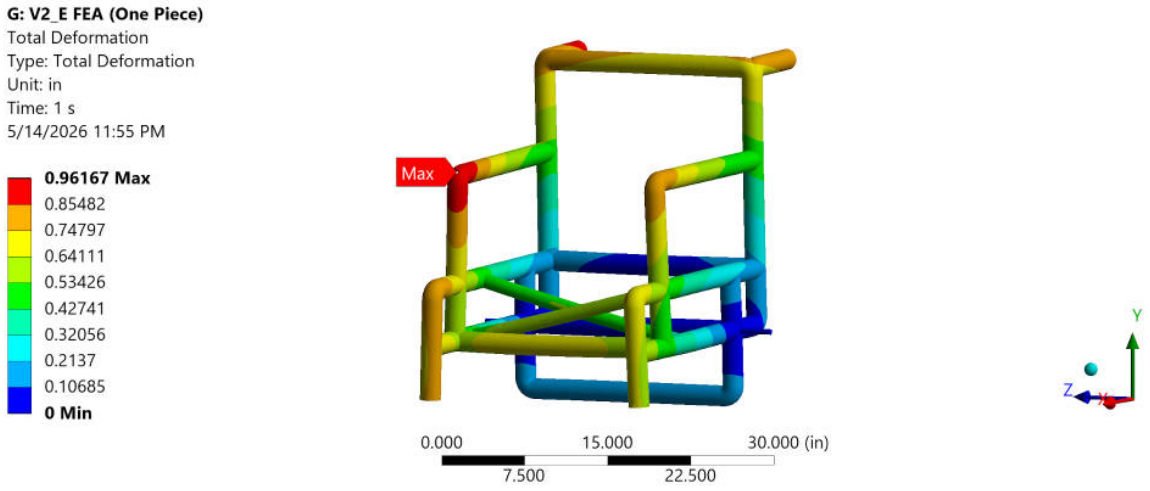


Figure 4.3.14: Total Deformation for Configuration E design for 200 lb load at mesh size 0.2 in

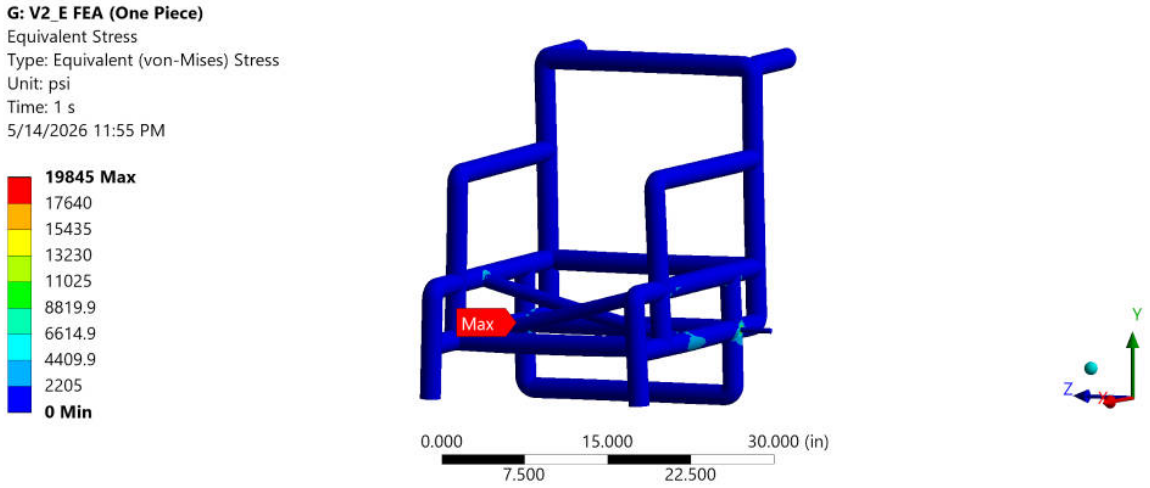


Figure 4.3.15: Equivalent Stress for Configuration E design for 200 lb load at mesh size 0.2 in

Based on the FEA model analysis, the configuration that we moved forward with, balancing structural integrity with comfort for the user, was Configuration E. It provided a good amount of structural support to allow for the total deformation of the chair to be minimal, while also providing increased comfort with the smaller pipes, a large criteria for the client's idea of an improved chair.

4.4 Prototypes and Testing

In choosing a mesh fabric, we followed the standardized testing described in AATCC TM162 to evaluate the material's chlorine resistance. Under pool manager supervision, we placed one sample of each material type in the pool for at least one hour. Throughout this hour, we agitated the samples while ensuring they remained in the water at all times. We then examined the visual and tactile differences between the chlorinated and untouched samples. We plan to test the tensile strength of the chlorinated samples versus the untested samples.

5 Detailed Design

5.1 Final Design

Overview

The final design of the Pool Accessible Wheelchair (PAW) is a highly durable, negatively buoyant mobility aid tailored specifically for aquatic therapy. The design retains the familiar form factor of a standard wheelchair but introduces critical structural and ergonomic upgrades to address previous issues of prolonged floating, backward tipping, component degradation, and user discomfort. The final assembly utilizes a heavy-duty Schedule 40 PVC frame, integrated steel ballasts, a relocated braking system, and a redesigned PVC/Olefin mesh seating arrangement. The final design is broken down into 4 subsystems: Frame, Mesh/ Fabric (Seat/ Backrest), Rear Wheels & Caster Wheels and Brake.



Figure 5.1.1: a) Final Design CAD **b)** Final Build (4th prototype)

Subsystem 1: Frame

The whole frame is constructed using 1-½" SCH 40 PVC (heavy duty PVC) that has its parts glued together by cement solvent or bolted. The cross support section is the only exception where 1" SCH 40 PVC is used.

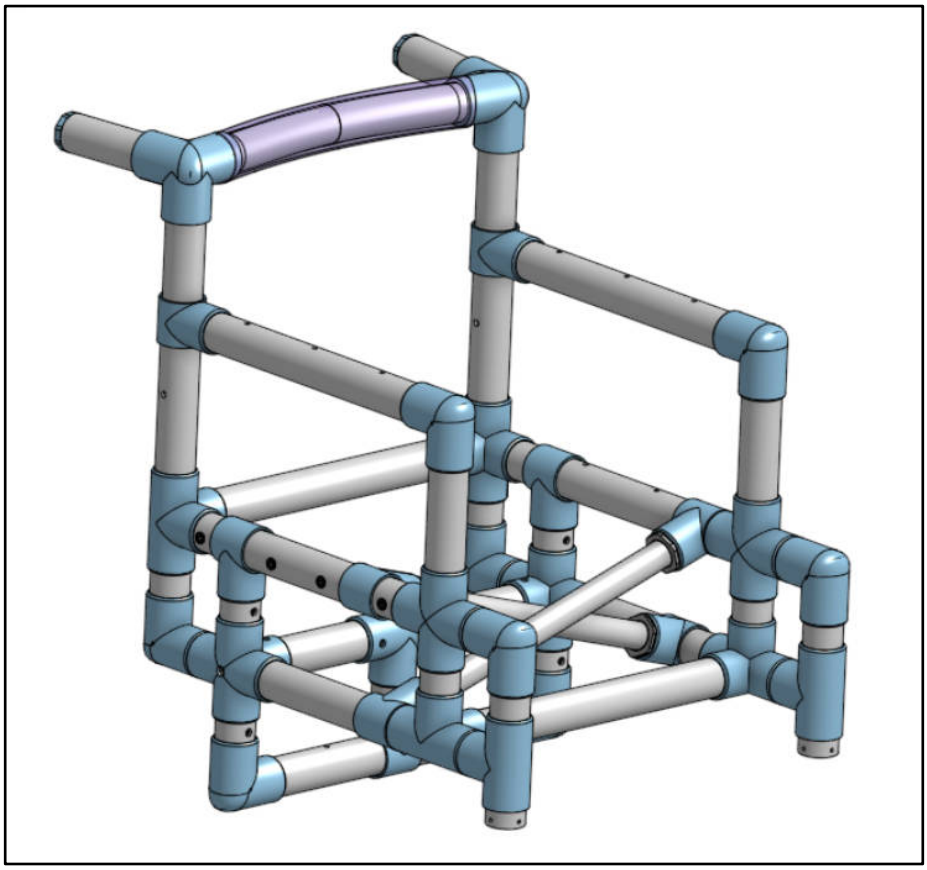


Figure 5.1.2: Frame CAD

To increase the comfort level, the horizontal pipe of the front seat is removed compared to the current PVC wheelchairs that are used at BAC. To ensure the structural integrity remained the same, a cross support system was specially designed in order to prevent the whole frame from caving in, which was the main issue that the team faced with the 1st and 2nd prototype. As mentioned, these cross sections are 1" SCH 40 PVC pipe connected with the main frame through SCH 40 1" x 1-½" Reducing Bushers and 1-½" SCH 40 Sling Tee (see Figure 5.1.2). The reason for using a smaller size of pipes here is to reduce the possibility of hitting the user's thighs due to the inevitably sagging issue of the mesh seat.

Along the frame sides, we use ¼"-20 Stainless Steel Rivet Nuts for mesh connection. The intention for this design is to move the screw location to the side of the frame where they

cannot rub against the user's legs, especially with people with lower body disabilities, any uncomfortable objects that create pressure could easily cause pain and even bruise on their skin due to the lack of muscle.

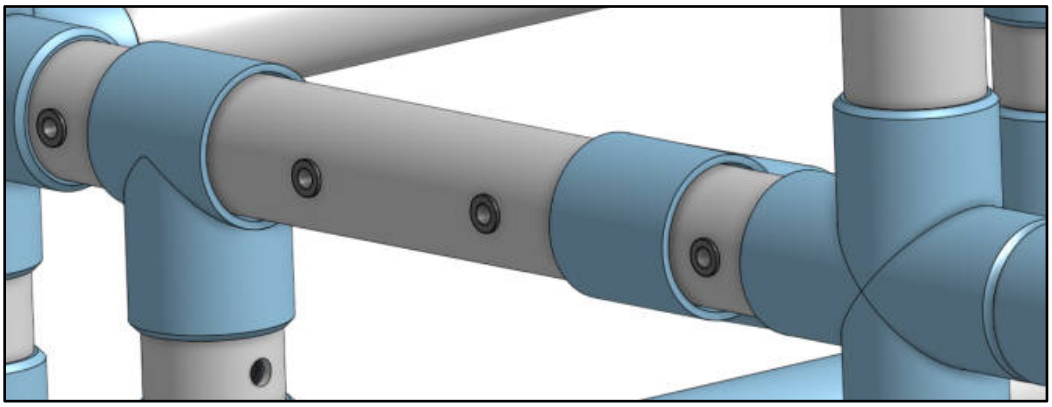


Figure 5.1.3: Rivet nut design for mesh connection

Vent holes of ¼” diameter are added along the armrests and the bottom pipe for water ingression, which helps reduce the buoyancy of the whole structure and contributes to resolving the tipping issue that happens with the PVC wheelchairs at BAC. (Figure 5.1.4)

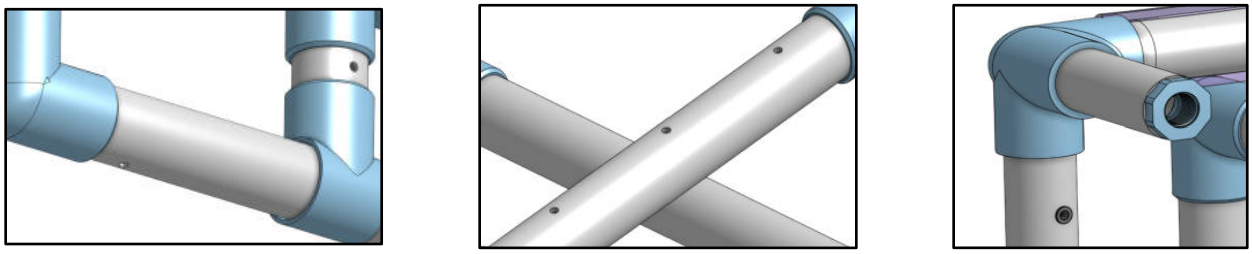


Figure 5.1.4: Vent Holes for Water Ingression

This curved backrest feature is kept as other PVC wheelchairs to increase the natural sitting posture, support spine, and also increase the comfort level overall.

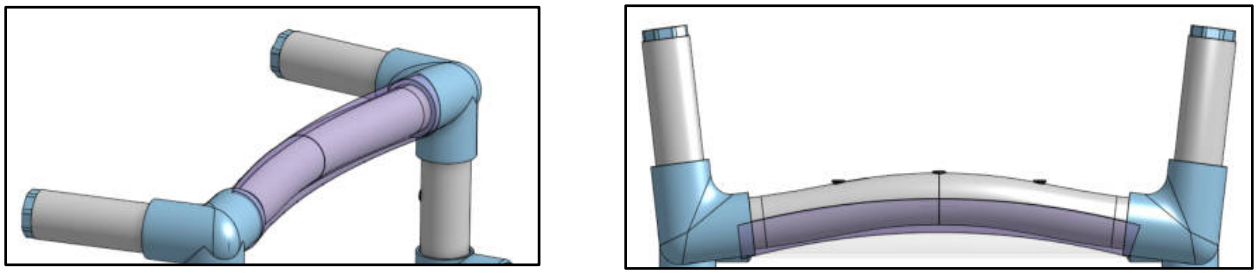


Figure 5.1.5: Curved Pipe for Backrest

Additionally, steel ballasts were inserted into the frame to increase the stability and reduce sinking time for the wheelchair. After testing different weights and location of ballasts (measuring the time it took to sink), it was decided that 2 x 2.5lbs steel ballasts placed in the front of the frame above the caster wheels was the most efficient without going overboard with the weights.

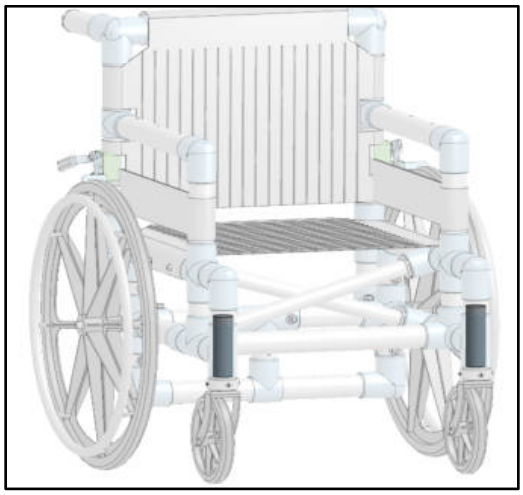


Figure 5.1.6: Internal Steel Ballast

The isometric exploded view of the whole frame structure is provided in Figure 5.1.7 below for assembly purpose. Note that some parts in this assembly will be overlapped with other subsystems because they share the same connections or functions.

Note: *If pipes have the exact same length but are notated with different capital letters (e.g., A, B), it indicates their critical design features differ. For example, the 22-inch pipes feature two distinct designs: Design A is machined for the seat frame area, while Design B is machined for the rear wheel system.*

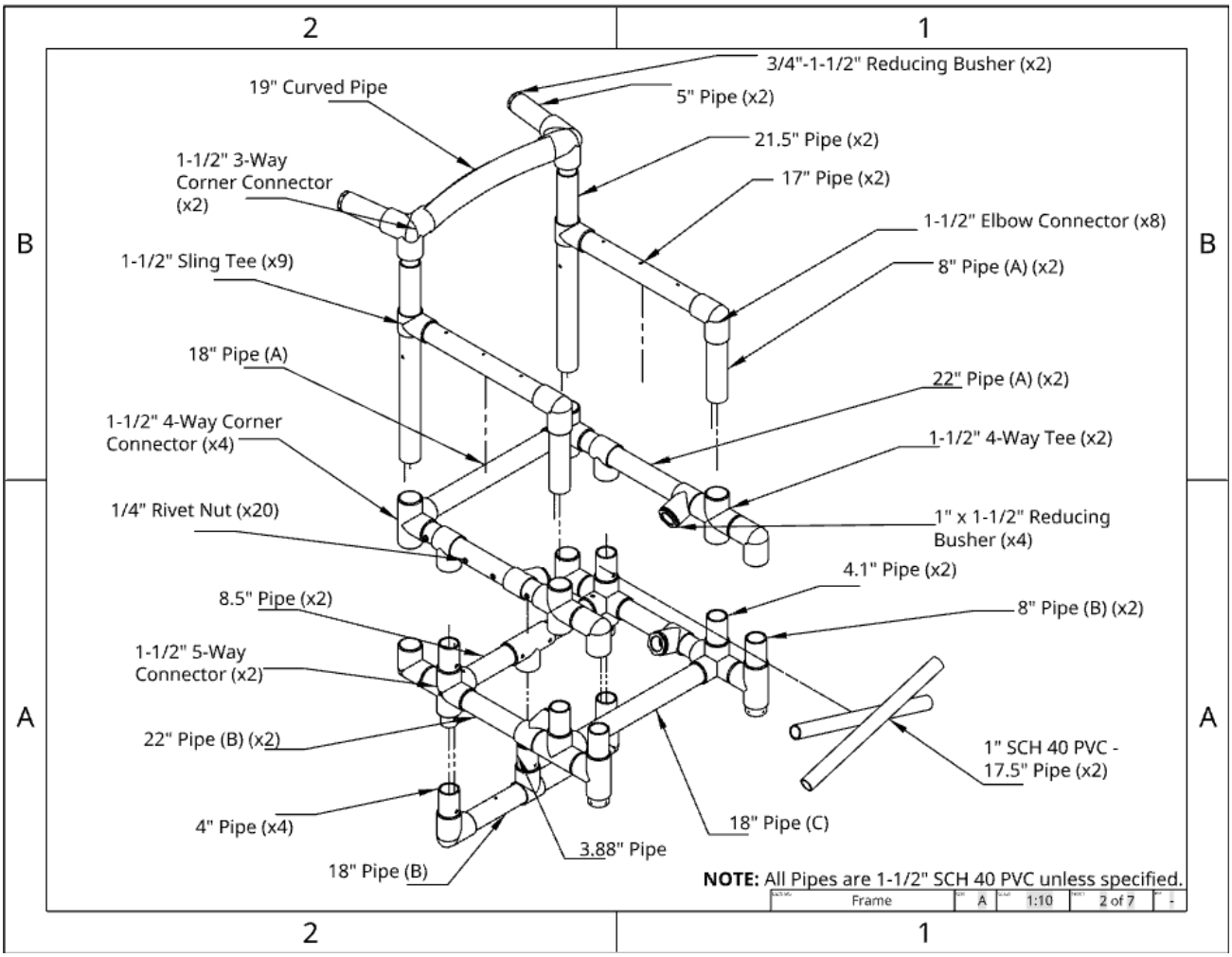


Figure 5.1.7: Exploded View of Frame for Assembly Purpose

Subsystem 2: Mesh/ Fabric (Seat/ Backrest)

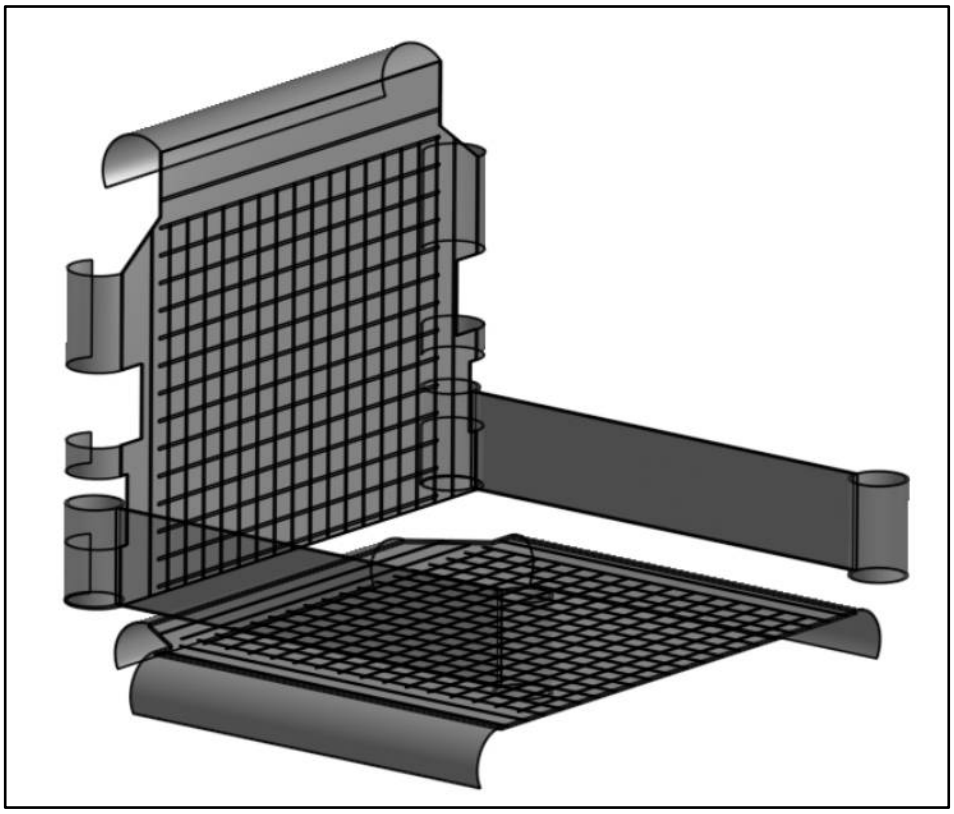


Figure 5.1.8: Mesh CAD

The seating and backrest utilize a woven vinyl-coated polyester and Olefin blend mesh (Phifertex PVC/Olefin Blend). Out of three different mesh blends, this material was selected after rigorous chlorine-resistance testing. Additionally, the materials were sampled for comfort using qualitative data by asking pool staff and visitors which one they would prefer to sit in. The final mesh was selected for its high dimensional stability to prevent sagging under the user's weight, an open weave for instant water drainage, and its comfort.

Originally, only one layer of mesh was applied to the chair in the V2 Prototype (Figure 5.1.9a), however, this was not strong enough to support the weight of the user. The mesh tore/stretched at the rivet nut locations. To account for this in the final prototype, the meshing was changed to have a total of 2 layers (see Figure 5.1.10). The first piece of fabric was folded on itself to make a thicker layer, then another one was placed on top of it with the meshing pattern facing 90 degrees from the first one, effectively creating a grid pattern that supported the strength of the overall mesh.



Figure 5.1.9: a) V2 Prototype Mesh **b)** V3 Prototype Mesh

As mentioned early in Figure 5.1.3, to address patient injuries caused by protruding hardware, the mounting method for the mesh was entirely redesigned. The mesh is now bolted directly into the *sides* and *back* of the chair frame, rather than the top edges of the seat. This removes all metal hardware from the patient's direct contact areas. Furthermore, mesh side panels were inserted along the armrest frames (Pictured in Figure 5.1.8b) to act as guards, preventing the patient's hands and clothing from rubbing against the rear wheels during transport.

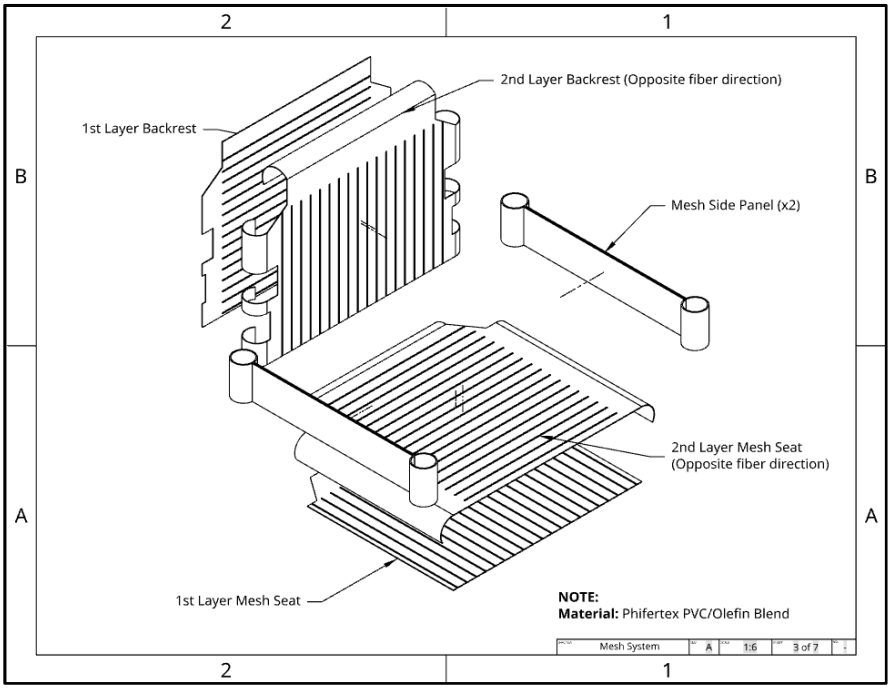


Figure 5.1.10: Layers of Mesh System

Subsystem 3: Rear Wheels & Caster Wheels

The mobility of the wheelchair relies on a specialized wheel configuration engineered to withstand and bear the weight of the entire chair and patient sitting on it. The rear drive wheels consist of 24-inch (we used a 23.5-inch wheel since that was what we had access to, but the design is based for a 24-inch wheel) mag wheels outfitted with airless tires. Mag wheels were selected over traditional metal-spoked wheels because their molded, single-unit construction eliminates the risk of spoke loosening, distortion, or localized galvanic corrosion at tension points under high cyclic loading. The integration of solid rubber, airless tires completely mitigates the hazard of pneumatic pressure loss due to puncture or long-term chemical degradation, ensuring a maintenance-free and reliable operational lifespan.

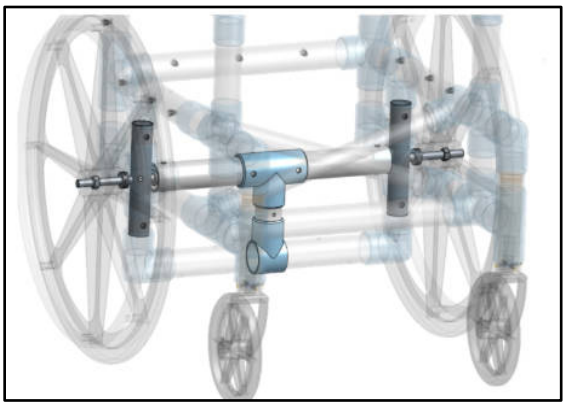


Figure 5.1.11: Rear Wheels System

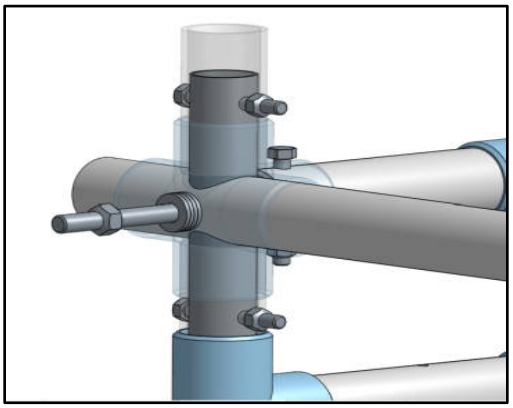


Figure 5.1.12: Rear Wheel Connector

Additionally, the center 1-½" Tee is used to connect two 8.5" horizontal pipes. This pipe is split in half and then connected with a tee to allow easy access to the ½" x 5" Steel Hex Screw that holds the tire together, making it simple to take the tires for maintenance. The entire rear wheel assembly is shown in the drawing below (Figure 5.1.13).

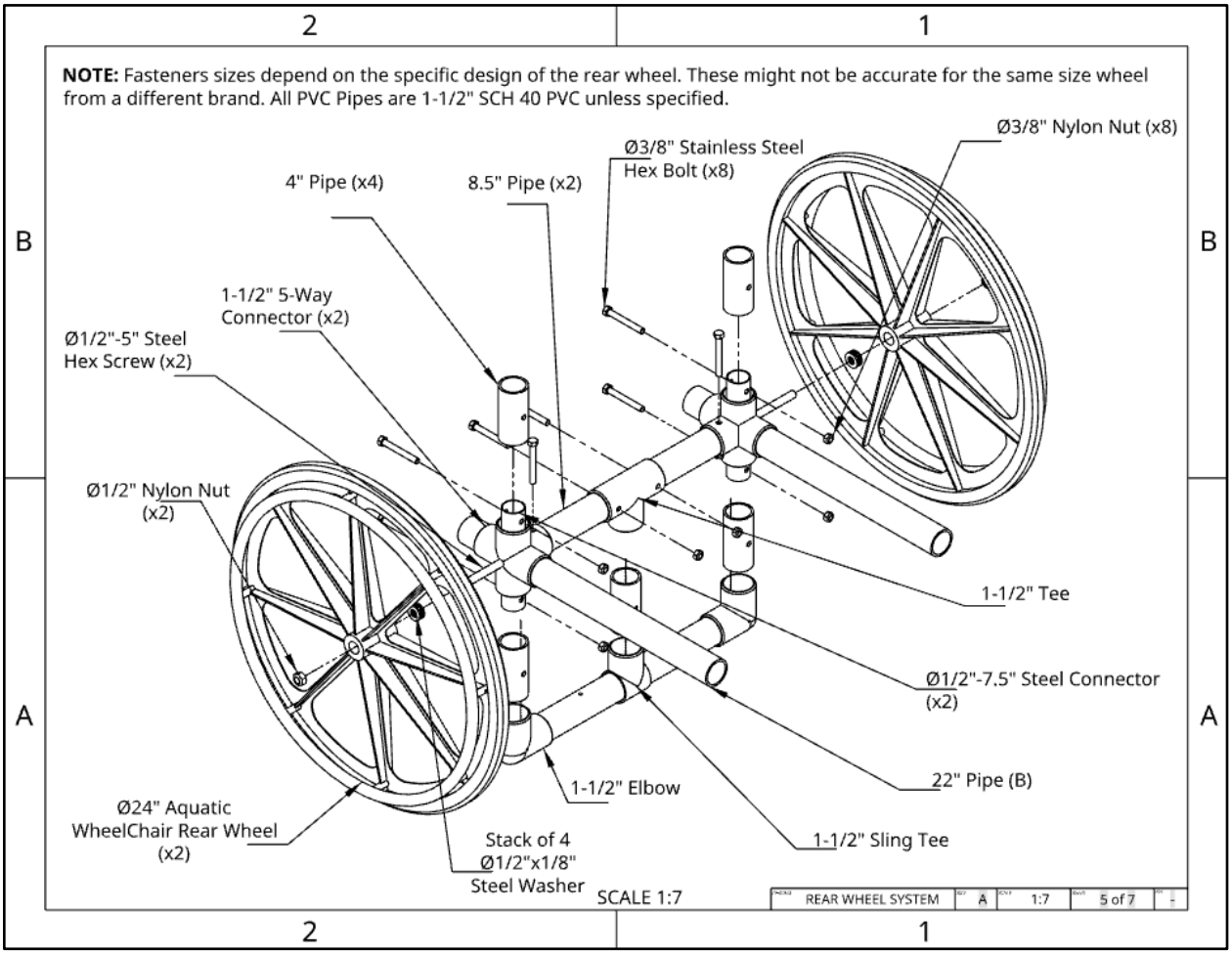


Figure 5.1.13: Exploded View of Rear Wheel System

For the front steering mechanism, dual heavy-duty caster wheels are mounted to the lower vertical PVC columns. As mentioned in Figure 5.1.6, the internal steel ballast is inserted to counteract the tipping and decrease the sinking time of the system. Additionally, the connector was machined to screw into the caster wheel. The connector is made to interface the wheel with the rest of the frame. This is shown in figure 5.1.14b.

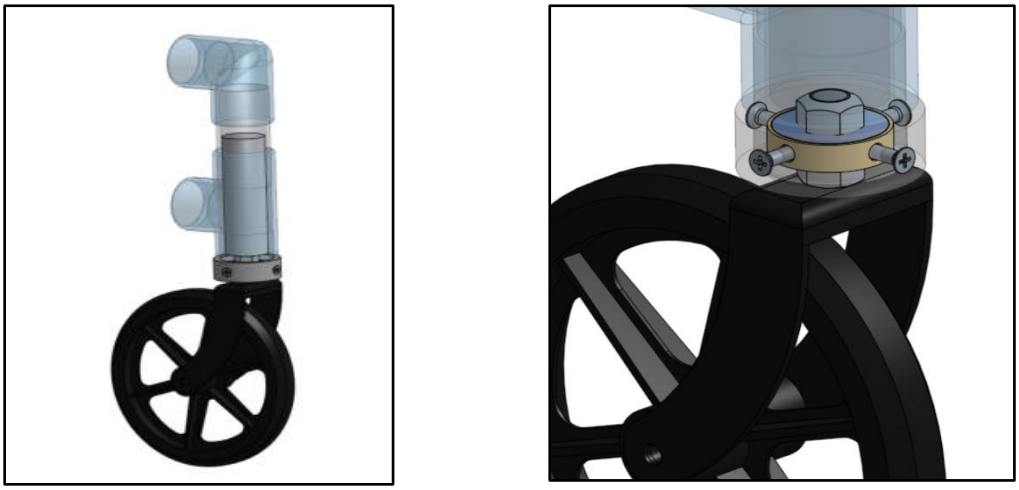


Figure 5.1.14: a) Caster Wheel CAD, b) Caster Wheel Connector

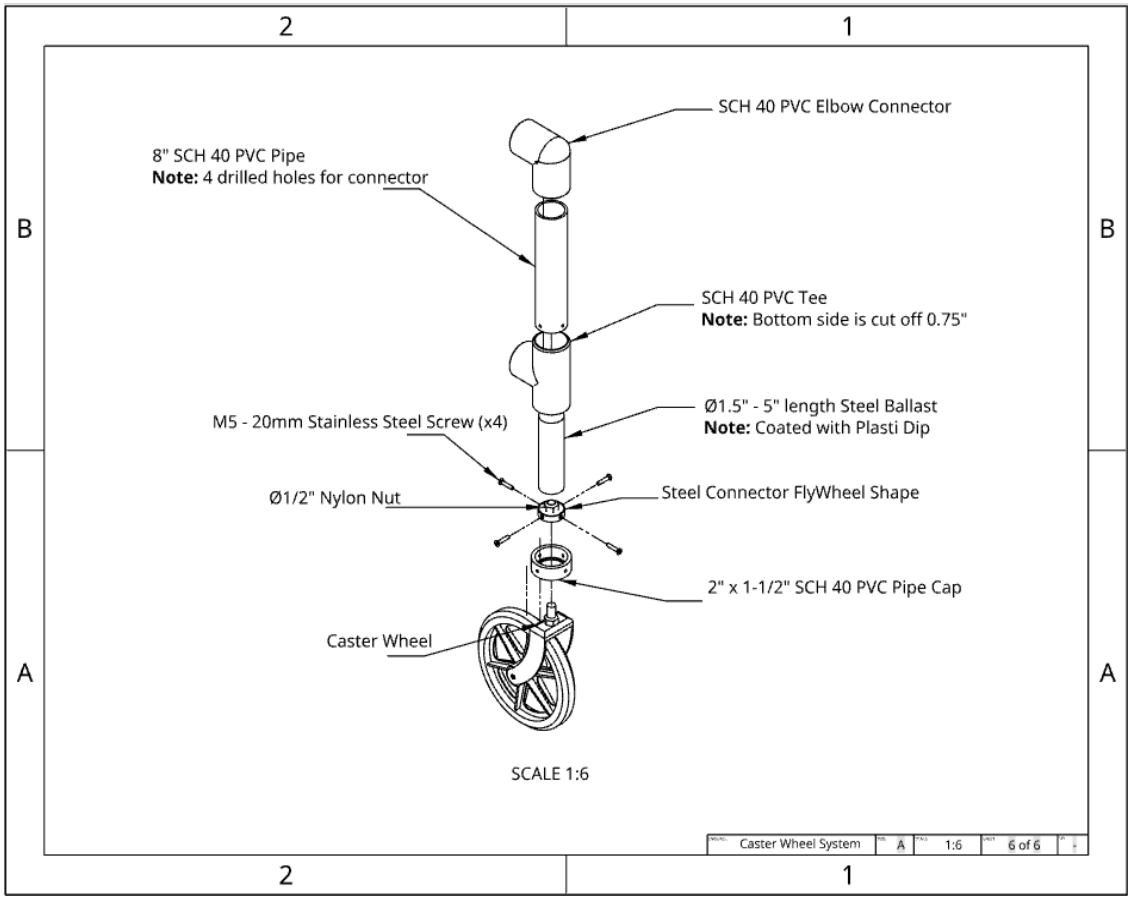


Figure: 5.1.15: Exploded View of Caster Wheel System for Assembly Purpose

Subsystem 4: Brake

The braking subsystem functions strictly as a stationary wheel-lock mechanism designed to hold the wheelchair completely stable on a standard 1:12 pool ramp while transferring the patient into and out of the chair. The final design utilizes a pull-to-lock brake configuration. In previous designs, forward-mounted toggle brakes were positioned within the patient's immediate reach, leading to frequent misuse as patients used the brake handles as leverage or grab bars to assist themselves during transfers. This repeated mechanical abuse caused rapid deformation of the mounting brackets and premature failure of the linkages.

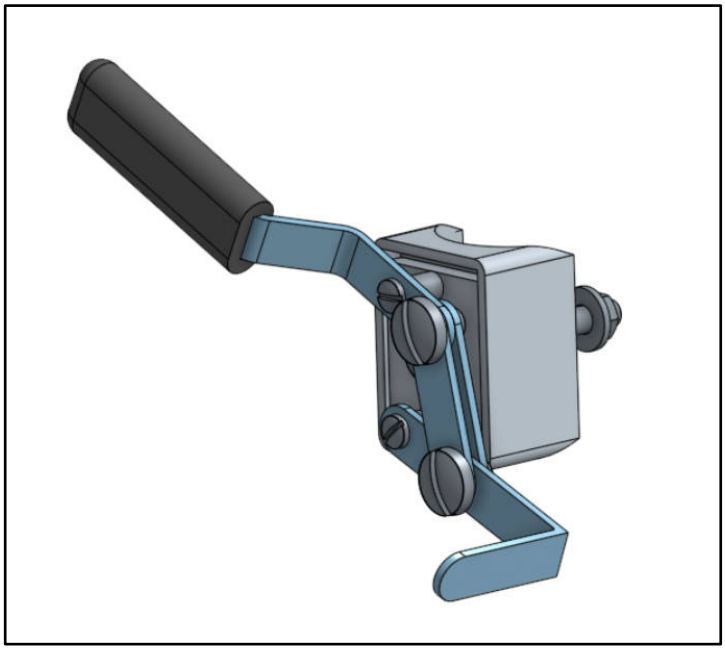


Figure 5.1.16: Brake System

To eliminate this human-error factor, the scissor brakes were completely relocated to a rear-mounted position on the frame, keeping the handle mechanism entirely out of the patient’s reach while remaining easily accessible at hip height for the physical therapist or operator. For the final prototype, we repurposed the stainless steel pool wheelchair’s brake, requiring us to design a new mounting bracket and update the brake from a horizontal braking system to a vertical braking system. The brake assembly is anchored to the PVC frame using a robust, multi-point contact mounting bracket. To protect the braking hardware from the warm, humid, and corrosive chlorinated atmosphere of the therapy pool, the 3D printed brake mount and brake connector was sprayed with Plasti Dip.

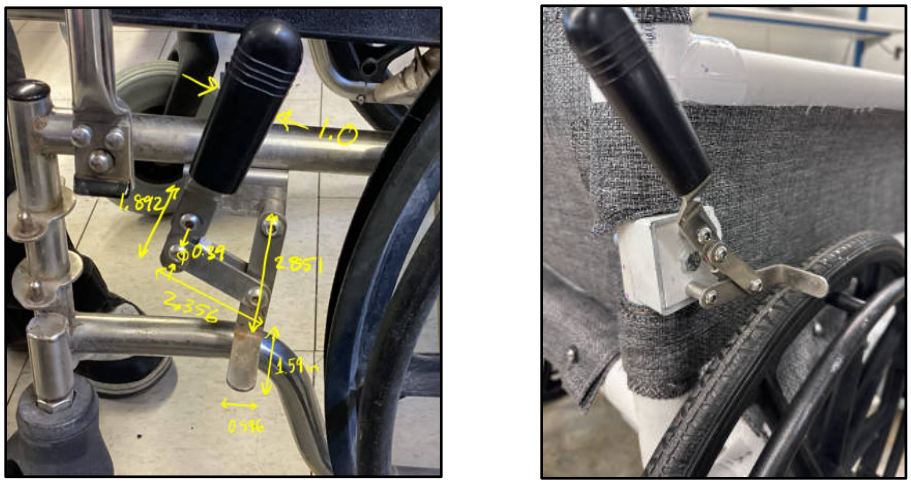


Figure 5.1.17: a) Horizontally-mounted stainless steel brake b) Final prototype vertically-mounted brake

When engaged, the scissor mechanism extends a high-friction brake bar directly against the solid rubber tread of the rear wheel, locking it in place. The geometric linkage of the scissor lock ensures an "over-center" mechanical advantage, meaning that the outward force exerted by the tire under load actually tightens the lock, preventing slipping or accidental disengagement even on wet, slick surfaces.

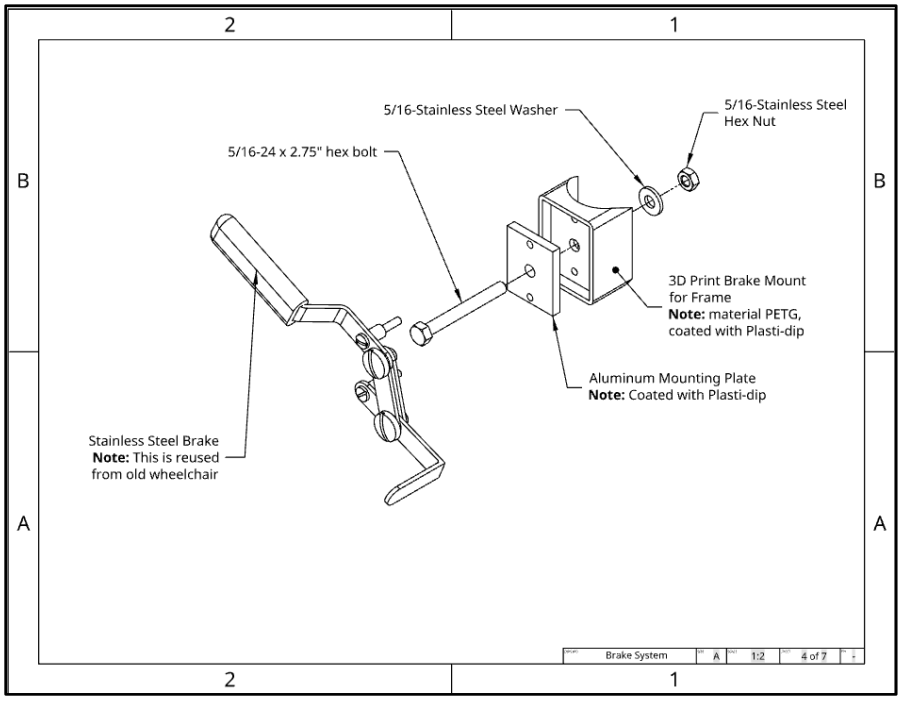


Figure 5.1.18: Exploded view of Brake System for Assembly Purpose

5.1 (B) Manufacture and Assembly

Manufacturers

The manufacturing and assembly of the design are detailed in the following technical drawings within this section, as well as in the Appendix of this document.

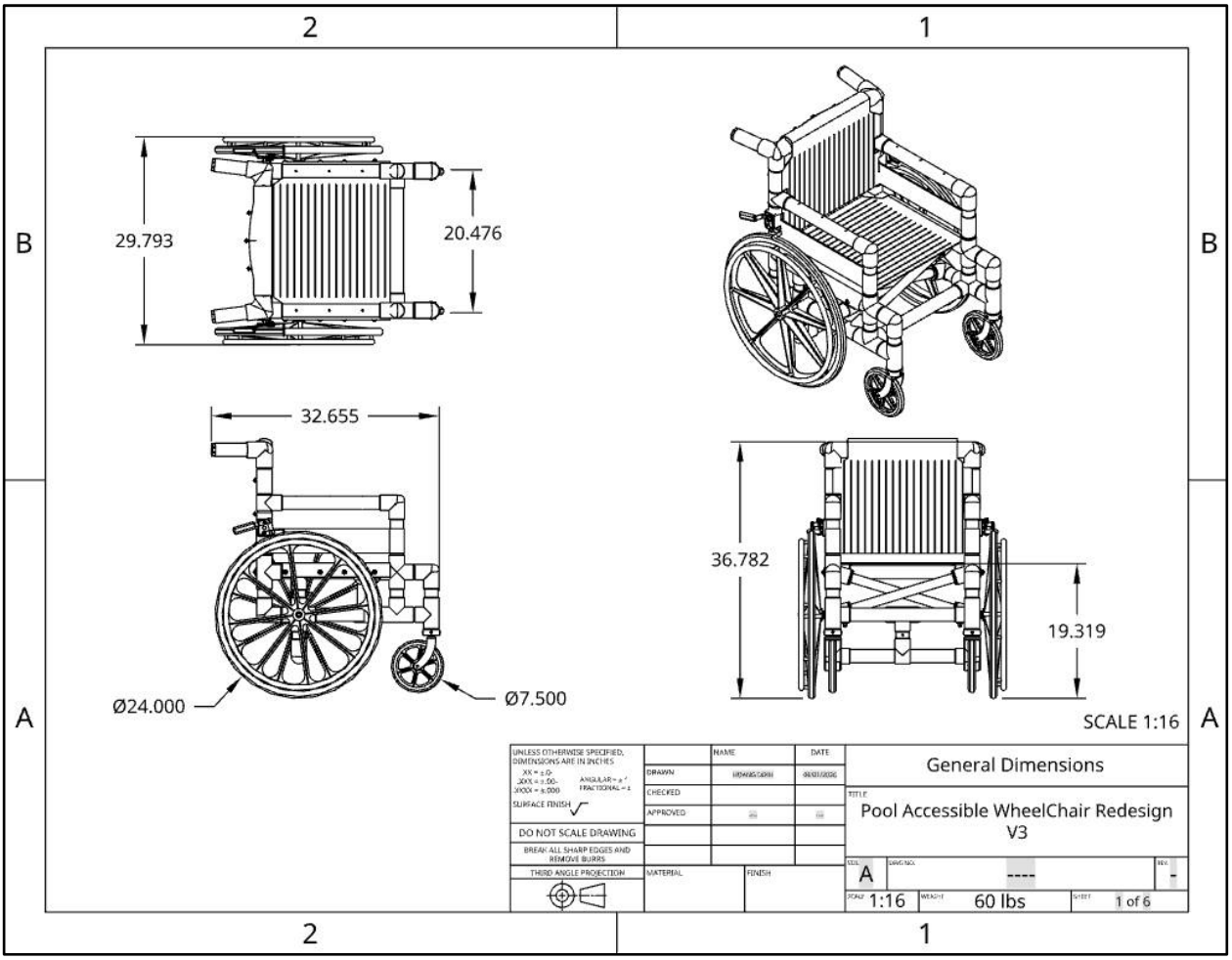


Figure 5.1.19: General Dimensions of The Final Prototype

An overall width of 29.793 inches ensures the chair can be easily pushed through standard doors and ramps at the BAC. The total height of 36.782 inches and the seat height of 19.319 inches provide a comfortable seating level for a wide range of users, while establishing an ergonomic handling height for therapists pushing the

wheelchair. A 24-inch outer diameter rear wheel and a 7.5-inch outer diameter caster wheel were selected, as these are industry standards for pool accessible wheelchairs.

In addition to the isometric assembly drawings of the four subsystems provided in the Overview section, this section details the critical features of the design. Please refer to the 2D technical drawings in the Appendix for specific part dimensions.

- **Pipes**: Must be cut perfectly straight. A horizontal drop-saw is highly recommended.
- **Connectors:** The inner diameter of all PVC connectors (excluding elbow connectors) must be machined to be approximately 0.015 to 0.020 inches larger than the outer diameter of the 1-1/2" Schedule 40 PVC. This creates a transition fit for easier assembly.
- **Vent holes:** ¼" diameter, and should be drilled after assembly.
- **Rivet Nuts holes:** 23/64" diameter, and should be drilled after assembly.
- **Metallic Parts:** Rear Rear wheel connectors and caster wheel connectors should be machined using a high-precision mill or lathe to ensure tight operational tolerances.
- **Mesh:** The entire fabric mesh system must be remeasured and fitted after the other three primary subsystems are built.
- **Curved Pipe:** To achieve the necessary curve for the backrest, hot sand (approximately 300–350°F) is poured inside the PVC pipe. The material will soften within one minute, allowing it to be safely bent to the desired radius
- **Fasteners:** All fasteners are recommended to be stainless steel.

Note: *If pipes have the exact same length but are notated with different capital letters (e.g., A, B), it indicates their critical design features differ. For example, the 22-inch pipes feature two distinct designs: Design A is machined for the seat frame area, while Design B is machined for the rear wheel system.*

Assembly

Once all components are manufactured and prepared, the assembly should be executed following these recommended steps:

Step 1: Frame Dry Fit and Marking

First, assemble all frame, brake, and wheel subsystems together without cement for a comprehensive dry-fit check. Once the general frame geometry is verified, clearly mark the alignment boundaries across every connector (see Figure 5.1.20 for reference). Marking these alignment redlines is critical; during the cementing process, parts must be pulled slightly apart to apply the solvent and then rapidly pushed back together to align perfectly with these marks before the cement cures.

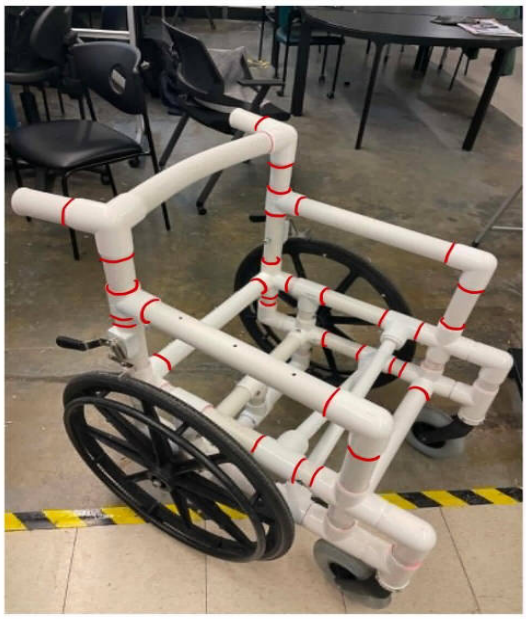


Figure 5.1.20: Marked Redline for Assembling and Cementing Processes

Step 2: Cementing Process

All PVC frame components must be permanently bonded using PVC cement solvent, unless they are explicitly designed to be mechanically fastened. The wheel subsystems (rear wheels and caster wheels) include components intentionally designed to be connected via hex bolts or screws. Accidentally cementing these dynamic parts will cause incorrect installation and compromise the wheelchair's functionality. .

For our team, the cementing process followed the two following figures:The cementing process should follow the sequential sub-assembly groups outlined in Figures 5.1.21 and 5.1.22.

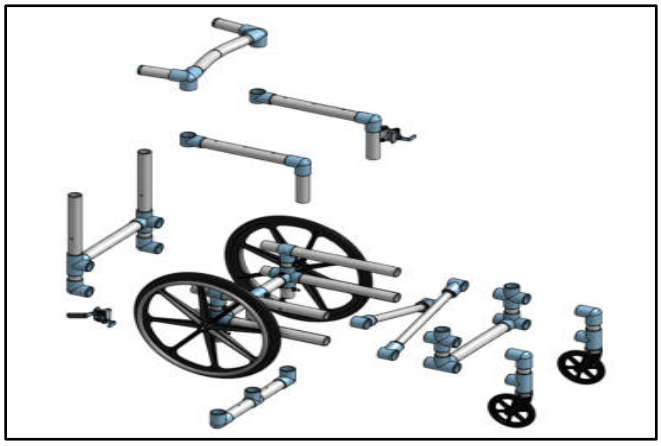


Figure 5.1.21: Exploded View of Each Group in Cementing Process

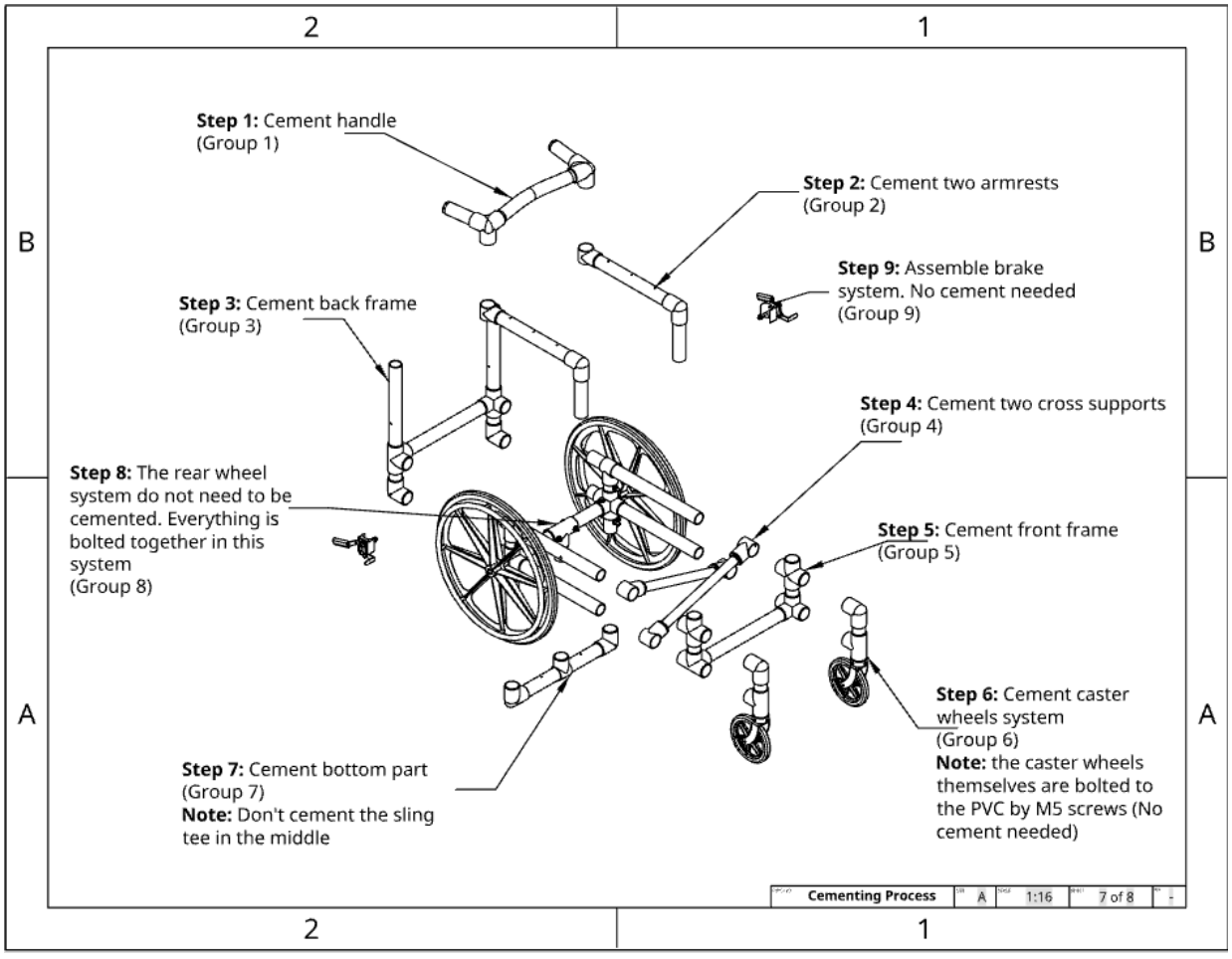


Figure 5.1.22: Cementing Process Step by Step

Step 3: Final Assembly

After all the sub-assembly groups are cemented and cured, begin the final integration using the red alignment marks established in Step 1. Taking the rear wheel system (Group 8) as the central reference base, slightly slide the back frame (Group 3) into the Group 8 fittings. Leave just enough space to apply the cement solvent, then firmly press them together to perfectly align the redlines.

Following this same technique, systematically connect Group 1, Group 4, Group 5, Group 6, and Group 7 into the main frame structure. Finally, bolt the assembled brake system (Group 9) onto the completed frame.

Note: *PVC cement solvent cures exceptionally fast. It is highly recommended to practice the application and alignment process on scrap PVC prior to assembling the final prototype.*

5.2 Engineering Economics

The primary economic objective of this project was to deliver a highly functional, corrosion-resistant aquatic wheelchair while strictly minimizing overall manufacturing and material costs. To achieve this, the design heavily leverages commercially designed components rather than expensive custom fabricated elements.

The most significant cost-saving measure was the selection of Heavy Duty Schedule 40 PVC for the primary frame architecture. During the initial material trade studies, standard 304 Stainless Steel tubing was evaluated but ultimately rejected due to its high material cost and complex, labor-intensive welding requirements. Schedule 40 PVC provides the necessary structural integrity (7,000 psi yield strength) and complete chlorine resistance at a fraction of the cost, approximately $1.44 per foot. Furthermore, by utilizing standard 24-inch rear wheels and 7.5-inch caster wheels, the team avoided the prohibitive costs associated with designing custom aquatic rated mobility hardware.

5.3 Budget and Costs

The average cost of a wheelchair that our clients typically buy, are between $1,200-$1,500 depending on the style and material. Our goal was to minimize the cost of materials as much as possible to then give room for the cost of manufacturing. By using PVC pipes, minimizing the usage of steel rods, and keeping track of the types of materials we use, we were able to make our wheelchair cost $900 in terms of materials, and with added costs of manufacturing, it would be about $1,100. The sealants and materials we used are also affordable and accessible, where shipping costs would not be an issue, except for the mesh for seating.

6 Verification and Validation

Our wheelchair was split into 4 different sections, frame, mesh, brakes, and wheels. So we each tested those aspects differently. In terms of verifying if our wheelchair was acceptable, the three main criteria we set were in regards to load capacity, sinking, comfort, and tipping. For load capacity, we wanted to ensure it would be able to withstand 350 lb load with less than 2 inches of caving. To ensure our wheelchair improved on sinking quickly, we aimed for the wheelchair to be fully submerged and touching the pool floor in less than 30 seconds. In regards to both comfortability and tipping, we set an 80% satisfaction rate for anyone willing to test it and compare it to the current wheelchairs in use. We started off by 3D drawing the original wheelchair our clients used, to be familiar with the dimensions of the wheelchair and which aspects to keep. When we finished that drawing, we then started calculating the rotational inertias and what was needed to keep the wheelchair from floating and tipping backwards. Once we had our calculation, we then decided on which materials to use, and decided if we used a heavier material then it would easily add

to the weight of the wheelchair without further expenses. We brought up the idea to our clients and they were very open and optimistic about those results, which encouraged us to continue with the decision. In terms of redesigning the frame for sinking time, we approached it with a general idea of inserting steel ballasts in certain parts of the wheelchair to prevent it from tipping backwards. To decide where to place the steel rods and how much weight, we performed multiple in person tests by bringing it into the pool, and timing its sinking time as well as taking turns sitting in the wheelchair to see how it would hold if a patient were to sit on it. We also took turns trying to tip the wheelchair by launching ourselves backwards while sitting in the wheelchair. While taking turns to sit on the wheelchair, we were able to visually see how the frame was upholding our client when they sat in it. We saw a pattern of it caving inwards a lot due to not having a middle support to keep the wheelchair extended, so we took inspiration from another wheelchair that our clients’ patients really liked, and decided to put a criss-cross sectional middle support bar to keep the wheelchair in place and also cemented all the PVC pipes to keep them from turning and slipping.

In terms of deciding on mesh, we researched on the types of materials for meshes, and agreed that PVC Olefin blend would be the most durable in chlorinated water as well as comfortable. So we grabbed 3 different samples that we thought would best be appropriate for the wheelchair and executed water tests in the pool such as leaving them in the chlorinated water for over a week to test degradation, swished the mesh around in the water for 3 hours and then set them to dry for 15 to see which mesh dried the fastest and which mesh tore if they did. And lastly, we allowed our clients’ patients to feel the mesh and take surveys on which mesh they liked best based on color and texture. With all of these results, we were able to choose a mesh.

Throughout our initial few prototypes of the wheelchair, our design was unsuccessful in meeting these marks, as the wheelchair frame had too much deformation and caving in to be considered safe for the user, and the sinking, while improved, did not reach our standard of less than 30 seconds. When users attempted to test the wheelchair’s comfort and tipping, users enjoyed the lack of a horizontal bar, but they were still able to tip the wheelchair, and noticed that they did not fully feel safe as well.

With our final prototyped wheelchair design, we were able to meet all 4 validation criteria that we set, confirmed by our clients through testing with them at the Bellevue Aquatic Center. Our total deflection in the design run through FEA came to 1.68 inches of maximum total deflection (shown below), and our sinking time for the chair dropped to 25 seconds. The user testing for comfort came to be an 87.5% satisfaction rating, while tipping success was 100% when users attempted to actively tip the wheelchair.

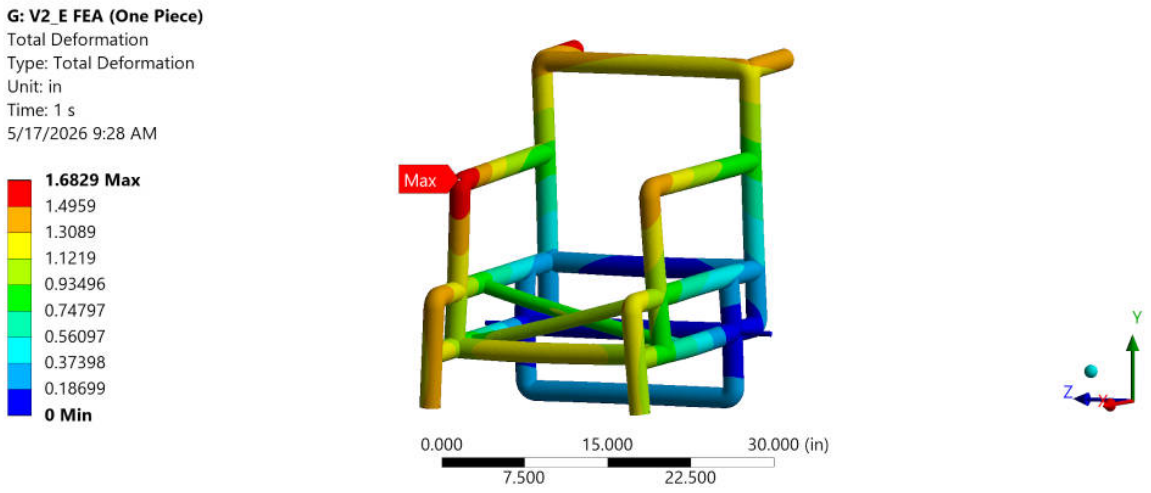


Figure 6.0.1: Total Deformation for Configuration E design for 350 lb load at mesh size 0.2 in

7 Discussion

7.1 Environment

Our wheelchair's lifecycle impacts occur during choice of materials (PVC, stainless steel, mesh, etc.), methods of manufacturing (cutting/drilling PVC, adhesives, coatings, etc.), use, and end-of-life. Our use of PVC pipes, stainless steel rods, and upgraded mesh material have their own energy emissions when created, and manufacturing the wheelchair design itself causes waste of cutting scraps and adhesive/sealing could add complexity to recycling. However, the biggest environmental impact comes during the use phase, due to its repeated exposure to chlorine and water, causing accelerated wear of the wheelchair parts.

With all these things in mind, we made multiple design decisions to help lessen the environmental impact, focusing on durability and repair to extend its lifecycle. We kept the mag wheel and scissor brake as necessary parts in our design, due to their robustness and low level of maintenance, two qualities that will greatly help extend the wheelchair's life. We upgraded the mesh material to be more comfortable and location to be more stable in use, reducing the need for mesh replacement. Stainless steel within the frame will help with frame structural integrity, increasing service life. When it comes time for the end-of-life process, we intend to disassemble and separate the materials for recycling. Although the steel rods and metal hardware will be easier to recycle, it is important to note that materials like PVC and the other polymer components are either limited in its recycling, so we will attempt to minimize mixed-material bonding to maximize materials recycled.

Wildlife interactions will be at a minimum, until the parts are recycled/replaced since it will be used in a controlled aquatic environment. For addressing additional environmental concerns, we recommend further research into chemical testing, its service life, and an end-of-life assessment, to determine how much our design actively extends its lifecycle and improves its environmental impact.

7.2 Safety and Liability

We have identified four main possible hazards that may arise during manufacturing and testing: lifting, collisions, human subjects, and water. All hazards will require us to ensure there is always at least one other person when working in the pool. We will also maintain necessary communication with the lifeguard on duty when at the pool.

The client prefers to move the wheelchair out of the water by reaching down from outside of the pool and pulling it out of the water. As of now, we plan for our final assembly to be less than 60 pounds, but the wheelchair is an awkward shape to pull out of the water. While we are working with the wheelchair and lifting it out of the water, we will follow the guidelines and proper technique demonstrated by the clients.

It is possible that the wheelchair assembly may wheel away and collide with others. It is important for those around the wheelchair to always be aware of it and its possible movements when in its proximity. When not using the wheelchair, the last person to use it will engage the brake/wheel lock system. The chair will be closely monitored during transportation and testing to prevent unintended movement.

Humans will be using the wheelchair and providing feedback. Those using the wheelchair may be at risk of slipping or losing balance. We plan to only allow those directly associated with the project (students, advisors, and clients) to fully test the wheelchair prototypes. Patients not associated with UW or the project may provide feedback based on appearance and materials in the form of an anonymous and optional survey. Patients will not sit in the designed wheelchair until it has gone through validation and verification by the project advisor, capstone professor, and clients. There will be controlled ramp use and slow, supervised transfer during testing.

All testing done outside of the capstone space will be conducted in the client’s pool at the Bellevue Aquatic Center. Testing will always be done with at least one facility member (pool manager, lifeguard, etc.) in the vicinity. All testing will follow the facility’s safety procedures and guidance from Wave Therapies staff and aquatic center personnel. All tests will be approved by the project advisor, capstone professor, and/or clients before being conducted.

7.3 Ethics

During iterating and testing the wheelchair, we must be sure that the patient can provide feedback while remaining anonymous to adhere to all parts of HIPAA. During final testing, we must look to gather feedback directly from the patients while not having them actually undergo human testing to avoid violating UW experiment guidelines.

7.4 Society

Since this project is aimed at serving people with disabilities, this design considers these things. Taking this into consideration, we designed the wheelchair to be easily accessible to those that use it. In terms of biases, when we identify the stakeholders to be the Bellevue Aquatic Center, persons with disabilities, and Aqua Creek (the manufacturer of the current chairs), we are able to look at different areas in our design that we could have been biased to. For the Bellevue Aquatic center, a few biases that we got from our assumptions are that the therapists are able to lift the chair out of the water without difficulty. Since this is what they typically do, we made the assumption that they are able to lift up the chair that we designed out of the water just as easily, however, we didn't consider how the extra weight on the wheelchair would affect their ability to do so. Another bias that we have is that everyone would be able to push the wheelchair. Since the chair handles are at one height, we assumed that everyone would be able to push it from that weight and have the strength to push the person in the wheelchair up the ramp. With this assumption, we also assumed that the trainer is trained and has the ability to handle people in wheelchairs. If this isn't the case, this can cause them to injure themselves and also put the person in the wheelchair at risk of falling out. The current design of the wheelchair does its best to address these biases, but it will require more testing. To test them, we would need the therapists to try pushing and pulling the wheelchair out of the water. Then, we can decide on design changes that we can make to accommodate the trainers to make it easier for them, such as fixing the handles.

For the people with disabilities that we are making the wheelchair for, there were a few biases and assumptions that we made. We made the assumption that they would be able to transfer themselves to the wheelchair and if they didn't, then we assumed that someone would be there to help them transfer. Under this assumption, anyone that needed to use the wheelchair would be able to use the wheelchair. However, if they can't transfer themselves and no one is there to help them, then they would not be able to use the wheelchair. This is something that we would have to consider in the final design, potentially designing the wheelchair in a way that is intuitive on how to transfer or design something in a way that makes it easier to transfer. Another assumption that we had was that we did our calculations for the design based on an average weight of people. Under this bias, this means that the wheelchair could either take longer or shorter times of sinking based on who is in the wheelchair. Although we designed the wheelchair to be

heavy enough to sink without anyone in it, it might take longer if someone is smaller or lighter. Since there is also only one size, this means that not all people could fit in the wheelchair. We did our best to address these biases with our design, making the weight heavier and moving the braking system out of the way so that it does not interfere with transferring, however, it would require physical testing to further get an idea of what improvements we could make to handle these biases.

7.5 Intellectual Property

The final presented wheelchair design will be the intellectual property of the city of Bellevue and Wave Therapies.

8 Conclusion

8.1 Summary

Our new wheelchair design serves as an improved PVC-based aquatic therapy wheelchair for Wave Therapies that upgrades many of the issues with the current design: lengthy periods of floating, durability, and uncomfortable contact points. The resulting design keeps many of the old elements and structure, but key changes consists of symmetric armrests, stainless steel rod inserts to increase weight and lessen floating time, an improved mesh material with better screw locations to remove unnecessary pain points for both the patient and the therapist, the continued use of mag wheels and a scissor brake due to its robustness and durability, and the brake moved towards the back of the wheelchair to prevent brake failure and misuse.

This design highlights the use of accessible materials, improved comfort and safety at many of the contact points, and upgraded durability by changing brake and wheel locations. The main weaknesses and things to consider are less than ideal materials used due to high costs and manufacturing constraints, and creating a design with the best possible weight to sink, but not be too heavy to take out of the water. Overall, the design is a practical, open-source-ready project improving many different aspects of the user experience while maintaining low cost and serviceability, but there are multiple characteristics that can be refined with additional testing.

8.2 Recommendation & Next Steps

The work moving forward will mainly come down to finalizing detailed design aspects, manufacturing our first prototype, and completing repeatable performance testing will help improve our design needs when it comes to strength, weight, durability, and comfort. Regarding design, through the prototype we will test the amount of steel and its locations, as well as the locations of the wheels and handbrakes and its effect on durability. To this end, we will also create an open-source manufacturing plan including a finalized BOM, cut list and assembly instructions.

For testing, our validation plan will focus on (1) buoyancy behavior of the wheelchair in the pool under realistic loading, (2) stability and tip resistance when the patient exists the wheelchair and it is fully submerged, (3) brake performance and durability due to its new location, and (4) the user and therapist experience when using and handling the wheelchair, checking for mesh comfort and screw location satisfaction through user surveys.

In addition to these, we plan to do safety testing on the wheelchair once the prototype is completed, which will mainly be tested by the client. This is to ensure that there are no liability issues with real patient testing. Before this though, we plan to validate the design with FEA, surveys, and look at previous incident reports to see how we could prevent those.

For next quarter, we plan to follow the gantt chart that we made. This includes finishing the prototype by 04/05/2026. Then we plan to finish testing and analysis by 04/19/2026. After we get feedback on our design, we plan to work on the beta prototype by 05/03/2026, then test it by 05/10/2026. Finally, we will use this feedback, wrap up the project with a final design, and do final testing for the capstone fair 06/03/2026.

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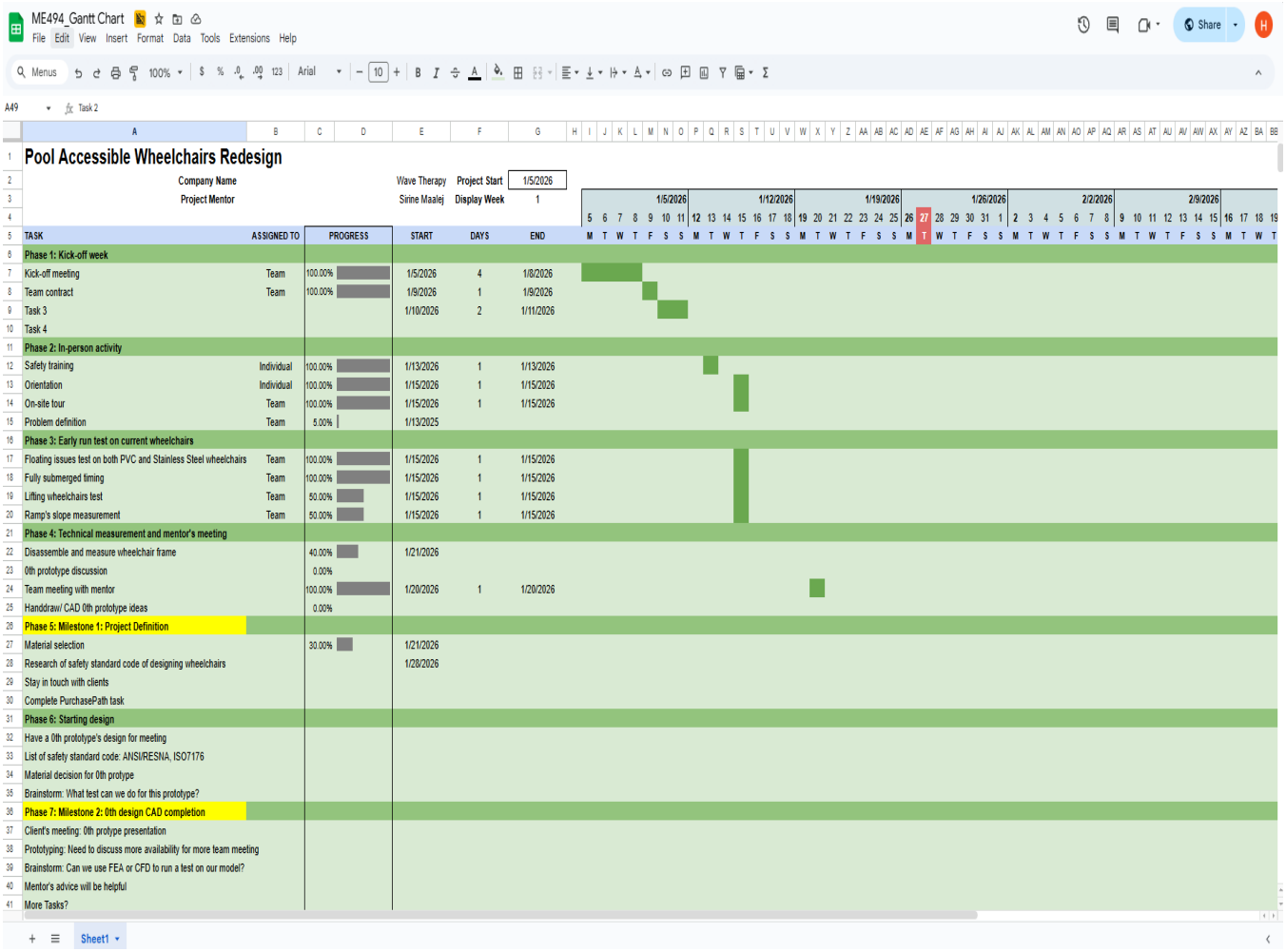
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Appendices

A Gantt chart [LINK]



ME494_Gantt Chart

File Edit View Insert Format Data Tools Extensions Help

Share

Pool Accessible Wheelchairs Redesign

Company Name			Wave Therapy	Project Start	1/5/2026
Project Mentor			Sirine Maalej	Display Week	1

TASK	ASSIGNED TO	PROGRESS	START	DAYS	END
Phase 1: Kick-off week					
Kick-off meeting	Team	100.00%	1/5/2026	4	1/8/2026
Team contract	Team	100.00%	1/9/2026	1	1/9/2026
Task 3			1/10/2026	2	1/11/2026
Task 4					
Phase 2: In-person activity					
Safety training	Individual	100.00%	1/13/2026	1	1/13/2026
Orientation	Individual	100.00%	1/15/2026	1	1/15/2026
On-site tour	Team	100.00%	1/15/2026	1	1/15/2026
Problem definition	Team	5.00%	1/13/2025		
Phase 3: Early run test on current wheelchairs					
Floating issues test on both PVC and Stainless Steel wheelchairs	Team	100.00%	1/15/2026	1	1/15/2026
Fully submerged timing	Team	100.00%	1/15/2026	1	1/15/2026
Lifting wheelchairs test	Team	50.00%	1/15/2026	1	1/15/2026
Ramp's slope measurement	Team	50.00%	1/15/2026	1	1/15/2026
Phase 4: Technical measurement and mentor's meeting					
Disassemble and measure wheelchair frame		40.00%	1/21/2026		
0th prototype discussion		0.00%			
Team meeting with mentor		100.00%	1/20/2026	1	1/20/2026
Handdraw/ CAD 0th prototype ideas		0.00%			
Phase 5: Milestone 1: Project Definition					
Material selection		30.00%	1/21/2026		
Research of safety standard code of designing wheelchairs			1/28/2026		
Stay in touch with clients					
Complete PurchasePath task					
Phase 6: Starting design					
Have a 0th prototype's design for meeting					
List of safety standard code: ANSI/RESNA, ISO7176					
Material decision for 0th protype					
Brainstorm: What test can we do for this prototype?					
Phase 7: Milestone 2: 0th design CAD completion					
Client's meeting: 0th protype presentation					
Prototyping: Need to discuss more availability for more team meeting					
Brainstorm: Can we use FEA or CFD to run a test on our model?					
Mentor's advice will be helpful					
More Tasks?					

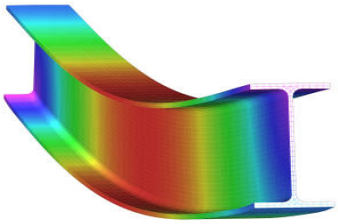
1/5/2026 1/12/2026 1/19/2026 1/26/2026 2/2/2026 2/9/2026

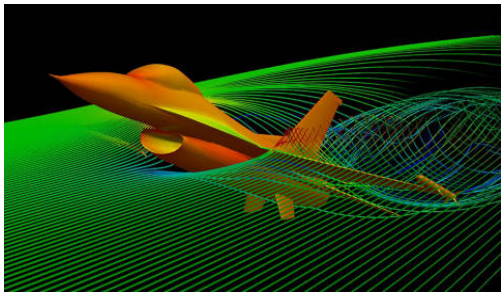
Sheet1

Figure A.1: Reference of Gantt Chart (*Please refer to the LINK to get access to the detailed timeline)*

B Verification and Validation Tests

Computational Testing (FEA)





Details:

- Structural analysis testing for new parts
- Test for stress, fatigue, loading
- FEA (Ansys)

Submersion Test



Details:

Phase A: Sinking Analysis

- Position the wheelchair at the water’s edge in a standard entry position
- Release the chair to initiate submersion and start timer
- Record visual data during the descent.
- Stop Timer when the chair rests firmly on the pool floor and all visible bubbling has ceased

Phase B: Draining Analysis:

- Vertically lift the half of the chair from the water to let the water drain + start timer
- Completely lift the chair from the water to the pool deck when most of the water is drained (Note the timer at this moment)
- Immediately weigh the chair now to compare it with the dry weight.

Load Capacity Test



Details:

- Based on the FEA result, slowly adding weight onto the wheelchair seat until it reaches the maximum load capacity from the analysis.
- Visually observe the deformation of the entire structure (mesh, frame, brake, wheels)
- Record any critical deformations and compare them with the FEA result to determine the recommended maximum load capacity for the design

Brake Test



Details:

- Testing using trials with clients to note:
  - Chairs stay still while the trainers move patients in and out of the chair
  - Brakes are out of the way so that patients do not use them as leverage

C Feasibility/ Analytical Calculations

LINK

Image is only for reference, please use the link above to access the detailed calculations.



Figure C.1: First page of the calculation PDF file

D Materials Specifications

A. Engineering Specs for Frame Summary:

	HDPE	PP	5052-H32 Aluminum	Heavy Duty PVC (Schedule 80 PVDF)	AISI Type 304 Stainless Steel	Heavy Duty PVC (Schedule 40 PVC)	Heavy Duty PVC (Schedule 80 PVC)
Yield Strength (psi)	2,390 - 4,600	4,060 - 4,930	28,000-33,000	6,500	31,200	7,000	7,200
Elastic modulus (ksi)	102 - 195	131	10,200	200 - 335	28,500	410	430
Chlorine Resistance	☑	☑		☑	☑	☑	☑
Specific Gravity	0.95 - 0.97	0.91	2.68	1.77 - 1.79	8.01	1.40	1.40
Cost (USD)	$0.89 / ft	$4.52 / ft	$20.93 - $80/ ft	$43.50 - $107.8/ ft	$61.93/6 ft	$1.44/ ft	$3.80/ ft
Nominal size	IPS/SDR 1"	1 ½"	1" OD	1 ½"	¾" OD	1 ½"	1 ½"

Figure D.1: Engineering Specifications for Frame (Milestone 2)

B. Engineering Specs for Mesh Summary:

Material	Yield Strength (psi)	Temperature range (degree F)	Elongation at break
(Vinyl) PVC Coated Polyester	1160 - 2900	-4 to 212	Avg 162%
Olefin Polymer	6670 - 9140	Up to 300	100 - 900%

Note: geometry and textile architecture (yarns' thickness and woven space distance) are the main factors to decide the purpose of different samples, even though they can be made of the same material

Samples	Main material	Dimension Stability	Chemical Resistance
Phifertex® PVC/Olefin Blends	Vinyl coated polyester + Olefin Polymer	High	Good
Phifertex® Plus	100% vinyl coated polyester	Excellent	Excellent
Phifertex® Cushion Collection	100% vinyl coated polyester	Low	Excellent

Figure D.2: Engineering Specifications for Mesh

C. Engineering Specs for Brake Summary:

	Central Caster Brake (Stroller)	Clevis Mount Brake	Newton Scissor Brake
Typical Static Hold (N)	2500	1800	2000
Operating Force (N)	90-130	40-65	35-50
Bending Resistance (N)	1315	600	400
Pivot Shear Strength (MPa)	190	205	280
Total Weight (g)	800-1200	200-350	150-250
Material Type	Stainless / Powder Steel	Aluminum / PVC / Stainless Steel	Carbon Fiber / Aluminum

Figure D.3: Engineering Specifications for Brake Selections

D. Engineering Specs for Wheel Summary:

	Mag Wheel (https://alcosales.com/al-72959-24-metal-mag-wheel/)	Spoked Wheel (https://sportaid.com/product/24-540-primo-sentinel-wheelchair-wheels-12-straight-pull-spokes/)
Wheel Diameter (in)	24	24
Tire Type	Solid Rubber	Solid Polyurethane
Hub Length (in)	2.25	1.875
Primary Materials	Black plastic wheel, chrome handrim	Double-wall aluminum rim + aluminum hub + anodized aluminum pushrim
Chlorine Resistance	Yes	Aluminum - Yes Spokes can rust
Load Rating (lb)	350	250-300

Figure D.4: Engineering Specifications for Wheel Selections

E 2D-Technical Drawings for Parts

LINK

Image is only for reference, please use the link above to access the details

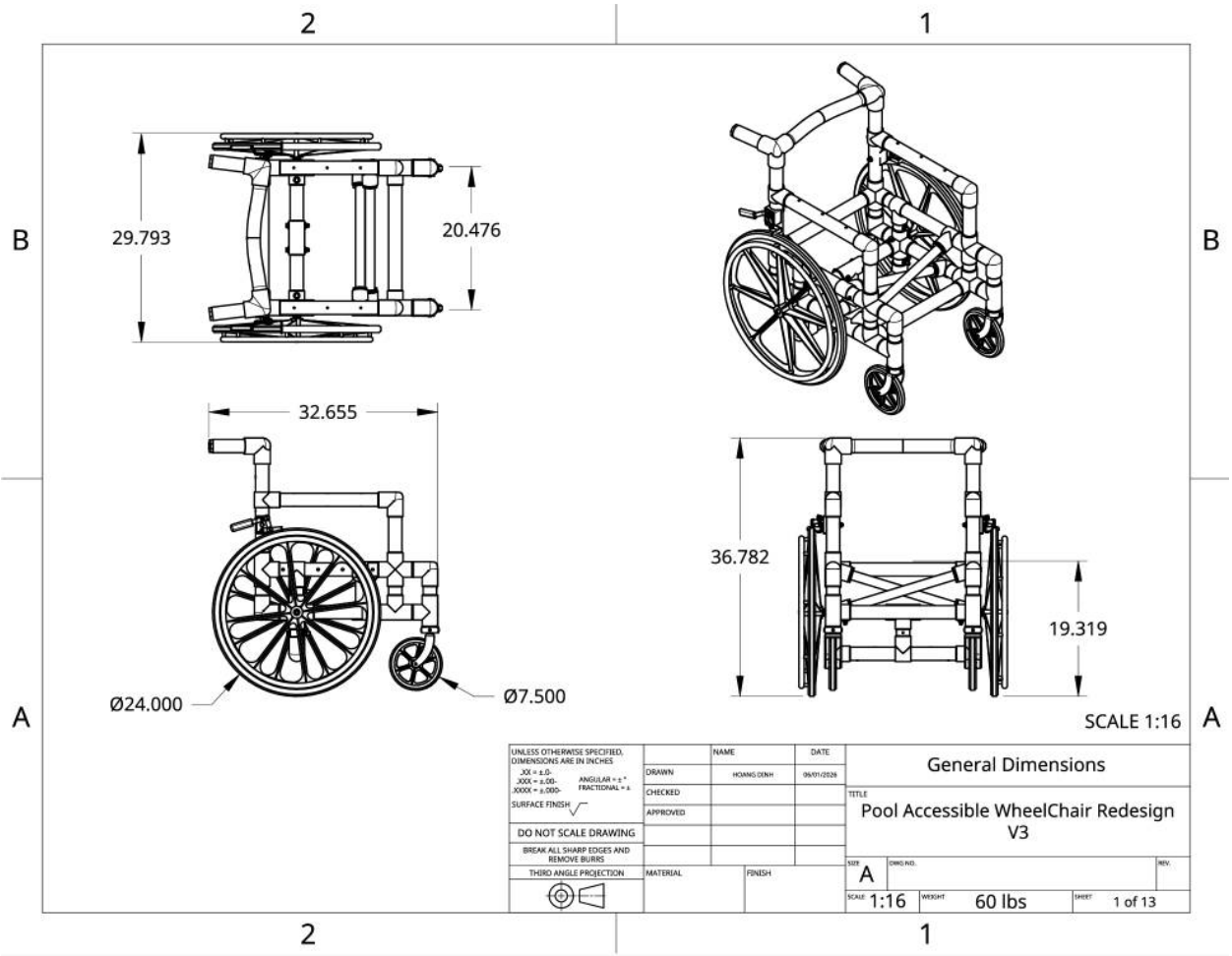


Figure E.1: V3 WheelChair General Dimensions

F 3D-CAD Models

LINK to CAD File

LINK to onshape File

Image is only for reference, please use the link above to access the details.

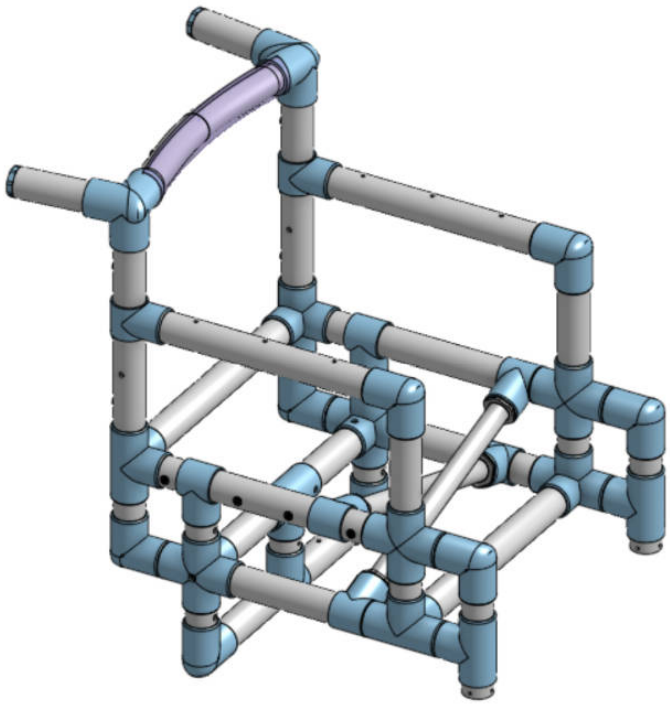


Figure F.1: V3 Frame Assembly as Visible in onshape File

G Finite Element Analysis

LINK

Image is only for reference, please use the link above to access the details.

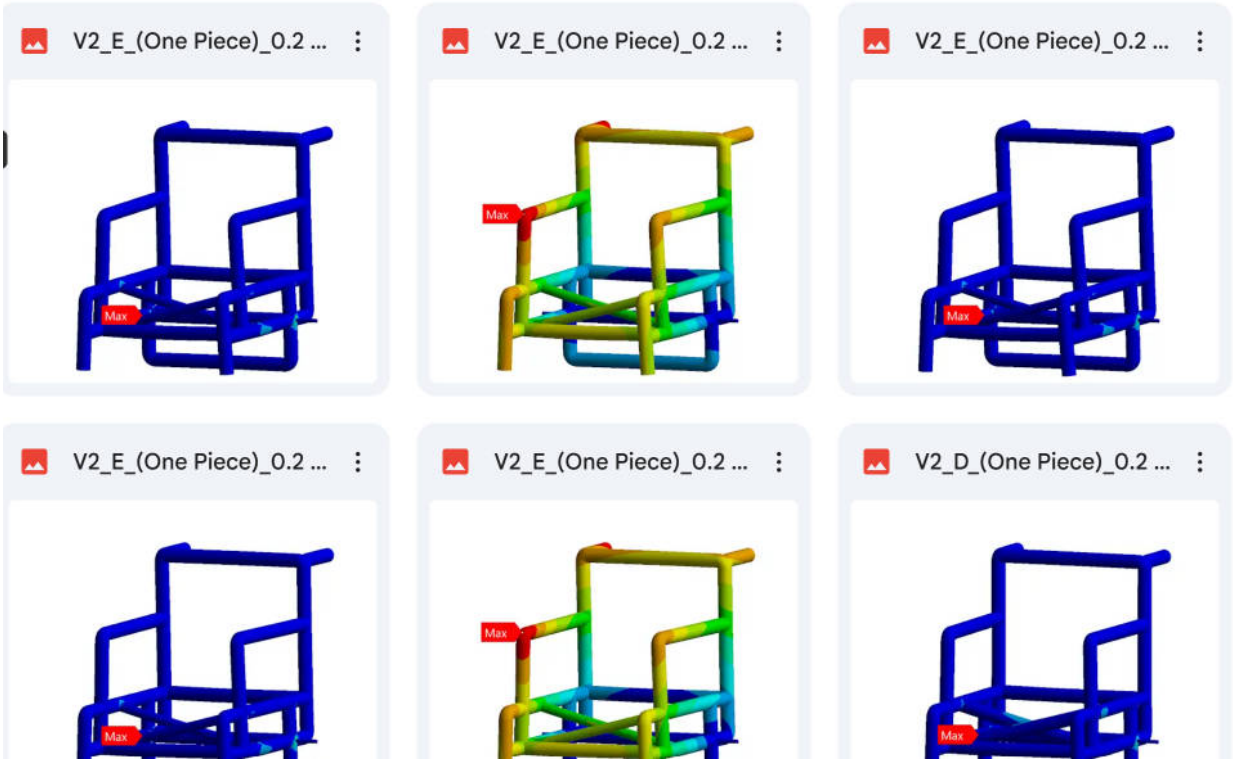


Figure G.1: FEA results as seen in the FEA file